

## REVIEW OPEN ACCESS

# Machine Learning Paradigm for Advanced Battery Electrolyte Development

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**Received:** 16 January 2026 | **Revised:** 28 February 2026 | **Accepted:** 20 March 2026

**Funding:** Scientific Research Innovation Capability Support Project for Young Faculty, Grant/Award Number: ZYGXQNJSKYCXNLZCXM-M4; National Outstanding Youth Science Fund, Grant/Award Number: 52222314; CNPC Innovation Found, Grant/Award Number: 2021DQ02-1001; Dalian Outstanding Youth Science and Technology Talent Project, Grant/Award Number: 2023RJ006; Dalian Science and Technology Innovation Project, Grant/Award Number: 2022JJ12GX022

**Keywords:** artificial intelligence | battery | deep learning | liquid electrolyte | machine learning | solid-state electrolyte

## ABSTRACT

As a crucial component of rechargeable batteries, electrolytes significantly determine interfacial characteristics and device performance. The development of advanced electrolytes relied on empirical trial-and-error and theoretical calculations mainly. Recently, the data-driven methods represented by machine learning (ML) have made progress in the materials field. However, interdisciplinary barriers and the multicomponent complexity of electrolyte systems impede the advancement of artificial intelligence (AI) in the electrolyte domain. Therefore, examining the application of ML methods in liquid and solid-state electrolytes to bridge the gap between battery science and data science is necessary. Firstly, this review summarizes the main technical routes and ion transport mechanisms of different electrolytes. Secondly, the basic concepts and suitable tasks of various ML algorithms are outlined. Following this, we investigate mainstream representation methods for electrolyte materials, which transfer material structures to machine-readable input vectors. Subsequently, representative application cases of ML in liquid and solid-state electrolytes are summarized. Finally, a perspective on the current challenges and future frontiers of AI-driven electrolyte research is provided. This review offers a deep understanding of this interdisciplinary field, providing intelligent insight for advanced battery electrolyte innovation.

## 1 | Introduction

Entering the 21st century, addressing climate challenges and achieving sustainable development goals through clean energy development is gaining widespread recognition [1–3], which drives advancements in the production, storage, and utilization of renewable energy [4–7]. Among energy storage technologies, electrochemical storage is one of the most important pathways [8]. Lithium-ion battery is the representative technology in this field, which has been integrated into most aspects of modern

life [9]. As the citation for the 2019 Nobel Prize in Chemistry states, lithium-ion batteries have created a world of rechargeable technology. Currently, many materials scientists are developing next-generation battery materials and novel battery systems with the aim of high energy density, long cycle life, and enhanced intrinsic safety [10–12].

Electrolytes serve as a critical component of battery materials, acting as a medium for ions transporting between the cathode and anode [13]. In addition, the properties of the electrolyte

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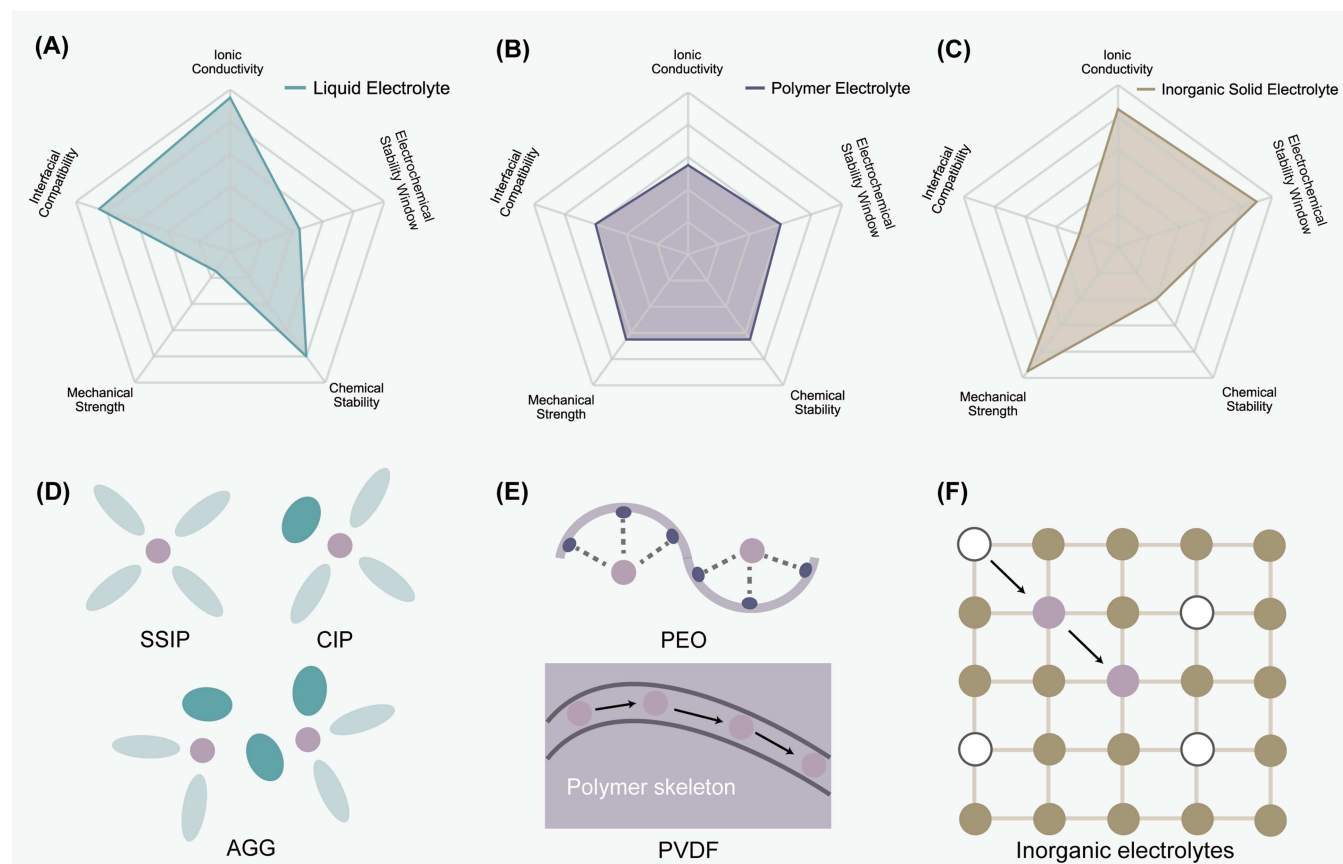
science. Therefore, we summarize frontier methods for material feature generation and structure representation. Following this, typical application cases of ML in the development of liquid and solid-state electrolytes were summarized. This review constructs a knowledge framework for the interdisciplinary field of electrolytes and ML. By summarizing key scientific issues and technical bottlenecks, it provides guidance for the intelligent development of next-generation battery electrolyte materials. Simultaneously, it provides methodological references for the AI-driven design for other material systems.

## 2 | Electrolytes

As a significant component of lithium batteries, the electrolyte undertakes the critical function of transporting ions between the cathode and anode to realize internal charge circulation. Multiple physicochemical properties, including ionic conductivity, lithium-ion transference number, thermal stability, and ESW, need synergistic optimization. High ionic conductivity reduces internal resistance and accelerates ion transport in the bulk phase, ensuring the high-rate operation of the battery. Meanwhile, a wide ESW elevates the energy density directly. Therefore, precise understanding and regulation of electrolyte chemistry is the key to constructing next-generation energy storage systems with high safety and longevity. As shown in Figure 2A–C, different properties of liquid and solid-state electrolytes are summarized and compared.

### 2.1 | Liquid Electrolytes

Liquid electrolytes are primarily composed of solvents, salts, and additives [41]. Carbonate solvents serve as the mainstream of commercial lithium-ion battery electrolyte. Cyclic carbonates contain ethylene carbonate (EC) and propylene carbonate (PC). Linear carbonates include dimethyl carbonate (DMC), ethyl methyl carbonate (EMC), and diethyl carbonate (DEC). Despite their strong polarity and high ionic conductivity, carbonate solvents form unstable SEI on lithium metal anodes. Ether solvents represented by 1,2-dimethoxyethane (DME) and 1,3-dioxolane (DOL) demonstrate superior affinity for lithium metal. Nevertheless, constrained by poor oxidation stability, ethers' electrochemical windows are typically below 4.0 V. Additionally, phosphate systems such as trimethyl phosphate (TMP) and triethyl phosphate (TEP) are attractive for their high safety attributed to their intrinsic non-flammability. However, they are unsuitable as primary solvents due to the lack of effective film-forming characteristics [42]. Apart from solvents, the selection of lithium salts considers solubility, electrochemical stability, thermal stability, synthesis accessibility, and toxicity.  $\text{LiPF}_6$ , LiTFSI, and LiFSI are all the main types of salts used. In addition, LiDFOB is typically used as a superior film-forming additive [43]. Moreover, the primary solvation structures influence ion migration behaviors while simultaneously determining the compositions of electrode/electrolyte interphases. As illustrated in Figure 2D, the solvation structures in liquid electrolytes are divided into three types, including solvation-separate ion pair (SSIP), contact-ion pair (CIP), and aggregate (AGG).



**FIGURE 2** | Properties summary and ion transfer mechanism of liquid and solid-state electrolytes. Radar plot of (A) Liquid electrolyte, (B) Polymer electrolyte, (C) Inorganic electrolytes. (D) Main solvation structures formed in liquid electrolyte, (E) Transfer mechanism in PEO and PVDF based polymer electrolyte, (F) Hopping mechanism of inorganic electrolytes.

and ionic aggregate (AGG) [44]. Consequently, introducing electron-withdrawing or donating groups [45], fluorination [46], and anion selection [47] impacts coordination ability and ion pair formation, enabling the regulation of electrolyte properties and battery performance.

## 2.2 | Solid-State Polymer Electrolytes

Solid-state electrolytes represent a current research hotspot in the battery field. They offer superior safety performance compared to liquid electrolytes. When coupled with a lithium metal anode, solid-state electrolytes can enable breakthroughs in energy density. Among them, solid-state polymer electrolyte (SPE) is an important technical route. Polyethylene oxide (PEO)-based systems serve as a mature choice. Here, Li ions continuously undergo dissociation-complexation with O atoms on chain segments within amorphous regions, thereby realizing transport in the bulk phase. Additionally, PVDF-based electrolytes represent another major route where flexible segments reduce local crystallinity to form fast Li<sup>+</sup> transport channels. Furthermore, Li<sup>+</sup> relies on residual DMF solvent to form dynamic coordination complexes. Continuous ion transport networks are established through interaction between solvent and polymer segments (Figure 2E). The advantages of the polymer route are simple preparation processes and excellent interfacial compatibility. However, the ionic conductivity at low temperature conditions is limited. Currently, strategies to regulate polymer ionic conductivity include polymer chain segment design [48], addition of plasticizers [49], compositing with inorganic solid particles [50, 51], and constructing novel localized high-concentration systems [52].

## 2.3 | Inorganic Solid-State Electrolytes

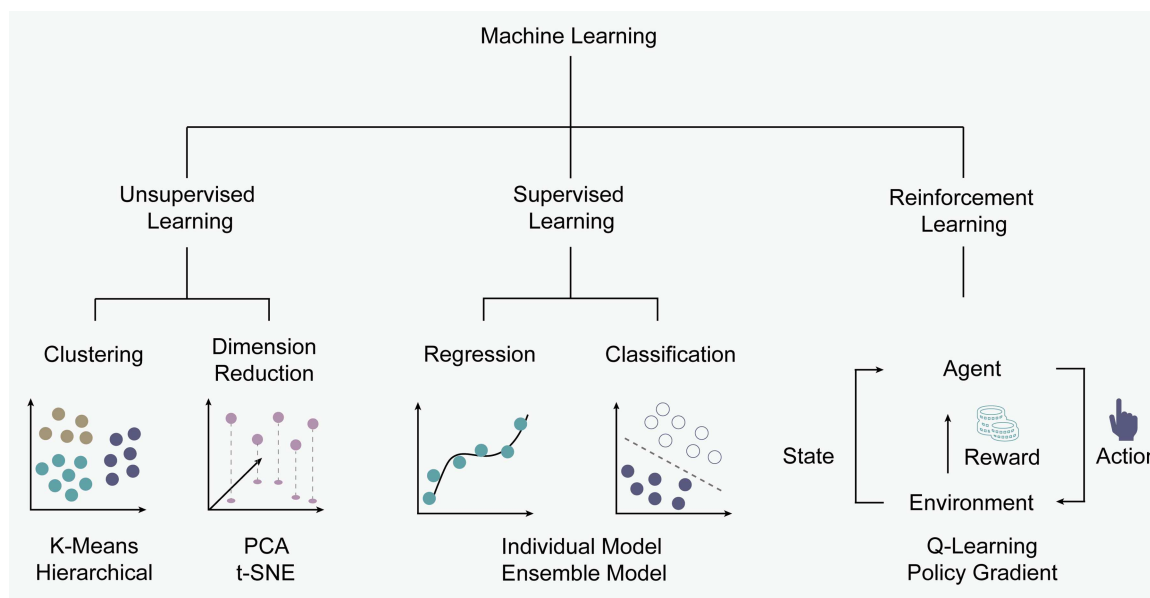
Inorganic solid electrolyte (ISE) is considered a key technical route for achieving intrinsically safe, high-energy-density

batteries [53]. Different from flexible polymers, ISEs utilize a rigid crystal lattice for ion transport via vacancy or interstitial hopping mechanisms (Figure 2F). Oxides and sulfides are the mainstream materials for ISEs. Among oxides, the Garnet-type (Li<sub>7</sub>La<sub>3</sub>Zr<sub>2</sub>O<sub>12</sub>) system offers wide ESW and favorable kinetic stability against lithium metal. Furthermore, sulfide-based electrolytes (e.g., LGPS and argyrodites) exhibit superior ion conductivity rivaling that of liquid electrolytes. This is attributed to the substitution of oxygen with highly polarizable sulfur anions, which creates expedient conduction channels. Additionally, halide electrolytes have emerged as a key route for high-voltage cathode compatibility due to their oxidative stability. Although ISEs enable superior ionic conductivity and mechanical modulus, their practical application is primarily constrained by poor solid-solid interfacial contact and chemo-mechanical instability. Strategies to regulate ISE performance include element doping [54], high-entropy design [55], buffer layer construction [56], and the development of composite electrolytes [57, 58].

## 3 | Machine Learning

### 3.1 | Basic Concepts of Machine Learning

ML aims to find underlying rules from data, recognizing complex patterns and making rational decisions. This review provides a summary of the fundamental classifications and algorithmic principles of ML, aiming to assist battery science researchers in rapidly establishing a knowledge framework. As illustrated in Figure 3, ML algorithms are often classified as supervised learning, unsupervised learning, and reinforcement learning. Requiring no preset labels, unsupervised learning reveals the intrinsic correlations of data through clustering analysis and feature dimensionality reduction. Representative clustering methods include the K-means algorithm and the DBSCAN algorithm. The former partitions the dataset into K compact clusters with minimized variance by iteratively



**FIGURE 3** | Overview of ML paradigms and representative algorithms. Main ML algorithms could be divided into unsupervised, supervised, and reinforcement learning.

optimizing the Euclidean distance between samples. And the latter identifies clustering structures of arbitrary shapes and automatically eliminates noise points based on density reachability. Principal Component Analysis (PCA) achieves dimensionality reduction by constructing a low-dimensional subspace composed of the top  $N$  orthogonal principal components, which capture the maximum variance in mean-centered data. This process enables linear reconstruction and information compression of high-dimensional features. After that, computational complexity was reduced while preserving core variability information. As a key component of ML, supervised learning constructs predictive models to learn the mapping relationship between inputs and outputs. Depending on the type of prediction objective, supervised learning can be further categorized into regression tasks for continuous values and classification tasks for discrete values. Firstly, we introduce a fundamental linear model of supervised learning. As shown in the following equation, linear models can be represented through a unified mathematical framework.

$$\hat{y} = X\beta = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p$$

Here,  $X$  denotes the data matrix, with  $\beta$  representing the coefficient vector. Driven by different regularization strategies, linear models have evolved from simple fitting to intelligent constraints.

$$L_{OLS} = \frac{1}{2n} \sum_{i=1}^n (y_i - \hat{y})^2$$

The coefficient matrix  $\beta$  was confirmed by minimizing the residual sum of squares  $L_{OLS}$ . Concise mathematical expression of a linear model reflects feature weights and ensures interpretability. At the same time, “structure–property” correlations in materials science offers direct design principle for material preparation. However, when excessive feature dimensions and redundant information appear, the traditional linear model is susceptible to overfitting.

$$L_{Lasso} = \frac{1}{2n} \sum_{i=1}^n (y_i - \hat{y})^2 + \lambda \sum_{j=1}^p |\beta_j|$$

Incorporating an  $L_1$  penalty term  $\beta_j$  into the loss function, Lasso regression was proposed. Lasso compresses some coefficients to zero, realizing sparse feature selection. However, the hard-thresholding characteristic of Lasso exhibits instability when handling highly correlated feature groups. Retaining only one feature within a feature group may lead to the loss of critical information of the dataset.

$$L_{Ridge} = \frac{1}{2n} \sum_{i=1}^n (y_i - \hat{y})^2 + \lambda \sum_{j=1}^p \beta_j^2$$

To address multicollinearity, ridge regression incorporates an  $L_2$  penalty term into its penalty function. Reducing the magnitude of coefficients rather than directly compressing them to zero, it mitigates multicollinearity issues and suppresses overfitting. However, ridge regression's soft thresholding strategy fails to

generate sparse solutions, thereby sacrificing interpretability of the results.

$$L_{ElasticNet} = \frac{1}{2n} \sum_{i=1}^n (y_i - \hat{y})^2 + \lambda_1 \sum_{j=1}^p |\beta_j| + \lambda_2 \sum_{j=1}^p \beta_j^2$$

Combining the advantages of both approaches, elastic net regression emerged. It employs  $L_1$  regularization for automatic variable selection while  $L_2$  regularization stabilizes coefficient estimates for correlated feature groups. This method ultimately balances the interpretability and fitting accuracy of linear models.

However, most of the material systems are multidimensional and nonlinear. So, nonlinear models play indispensable roles. Nonlinear models are primarily categorized into single models and ensemble learning models. Support vector machines (SVMs) map data to a high-dimensional space by kernel functions, then seek the optimal hyperplane within this space to accomplish regression and classification tasks. SVM models exhibit steady generalization capabilities in small sample scenarios, but increasing sample volume leads to higher training complexity. Unlike SVM, decision tree models explicitly construct tree structures to simulate human decision-making processes. A decision tree comprises a root node, several internal nodes, and multiple leaf nodes. The process of determining nodes and edges involves identifying optimal features and partitioning points to divide the data space into similar subregions.

Both SVM and decision trees are single models, showing limited generality under large-scale datasets. Hence, ensemble models integrate multiple base learners to enhance generalization performance. For example, Random Forest (RF) constructs many decision trees using bootstrap sampling and aggregates their predictions to conclude. This parallel ensemble strategy effectively reduces the variance and overfitting risks compared to individual models. Simultaneously, RF outputs feature importance rankings, enabling researchers to identify key material descriptors and providing scientific guidance for subsequent experimental design. However, the parallel mechanism of RF has limited effectiveness in reducing model bias. The Boosting algorithm addresses this issue by adopting a serial training strategy, wherein each newly generated tree focuses on correcting the residuals left by its predecessor. Specifically, XGBoost achieves precise minimization of the loss function by incorporating a second-order Taylor expansion into its objective function. Furthermore, it employs regularization terms to control model complexity, achieving an optimal balance between prediction accuracy and generalization capability. Moreover, LightGBM is deeply optimized for large-scale datasets. By utilizing histogram-based decision tree algorithms and a depth-restricted leaf growth strategy, it significantly improves computational efficiency while maintaining high predictive accuracy.

Reinforcement learning (RL) algorithms simulate the “trial-and-error” behavior in biological systems. This characteristic is suitable for sequential decision-making tasks in uncertain environments. Agent, environment, state, action, and reward are the core components of RL. At each time step, the agent observes the state of the environment, making decisions and

taking actions. Subsequently, the environment undergoes a state transition upon receiving the action and provides a reward signal. This closed-loop process of “observation–decision–execution–feedback” generates sequential data that forms the basis for policy evaluation. The objective of agents is not to pursue instantaneous rewards but to maximize long-term cumulative rewards by updating their policy. Thereby, RL achieves optimal control and adaptive evolution within complex dynamic environments. Representative algorithms include Q-learning and REINFORCE.

Q-Learning is a model-free and off-policy value learning algorithm. It evaluates the expected future cumulative discounted reward that can be obtained by executing an action in a specific state by updating the state–action value function. The rule is defined as follows:

$$Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

Here,  $\alpha$  denotes the learning rate controlling the speed at which new observations overwrite old values, while  $\gamma$  represents the discount factor balancing the importance of immediate rewards and long-term returns. The REINFORCE algorithm represents policy gradient methods and differs from the value function learning path of Q-Learning. It optimizes policy parameters via gradient ascent to maximize the expected total reward. If an action sequence yields a positive cumulative return  $G_t$ , the selection probability of state–action pairs within that sequence gets enhanced. Therefore, the policy gradient update direction is determined by the following equation:

$$\theta \leftarrow \theta + \alpha \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) G_t$$

$\nabla_{\theta} \log \pi_{\theta}(a_t | s_t)$  represents the logarithmic gradient of the policy, signifying the direction of parameter adjustment of  $a_t$ .  $G_t$  function is a crucial weighting factor that modulates the magnitude and sign of this update.

### 3.2 | Basic Concept of Deep Learning

Deep learning (DL) is a branch of ML with more parameters and more complicated structures. DL employs a multilayer artificial neural network (ANN) to simulate the hierarchical abstraction mechanisms of the human brain [59]. ANN autonomously extracts features directly from raw input data, bypassing the feature engineering steps of traditional ML [60]. This establishes an “end-to-end” mapping from input to output. ANN excels at processing high-dimensional complex data such as images, natural language, and graph structures [61]. This section provides a systematic overview of the foundational architecture of DL. The fundamental computational mechanism of artificial neurons is illustrated as follows:

$$y = f(w^T x + b)$$

$x$  represents the input vector,  $y$  represents the output vector,  $W$  represents the weights,  $b$  is the bias, and  $f$  denotes the activation function. Activation functions introduce nonlinearity into neural networks. A Feedforward neural network (FNN), also

named a multilayer perceptron (MLP), is constructed by stacking these neurons hierarchically. They consist of an input layer, hidden layers, and an output layer, propagating information in a unidirectional manner. The optimization of network parameters relies on the backpropagation algorithm. This method leverages the chain rule to efficiently compute the gradient of the loss function with respect to the weights and biases of each layer. By iteratively updating parameters in the direction opposite to the gradient, the network progressively approximates the target mapping.

However, MLP ignores the spatial structure of data as it only receives a 1D vector. Image data possesses a 3D structure containing height, width, and channel information. So, it must be flattened into a 1D vector and input for MLP. This data flattening approach disrupts spatial relationships within the data, neglecting the visual features such as local patterns and geometric structures. To overcome this limitation, a convolutional neural network (CNN) effectively extracts these spatial features by simulating the local perception mechanisms found in biological visual systems [62, 63]. A learnable convolutional kernel traverses the input space in a CNN. By performing weighted local sums, this kernel extracts the essential features of the 2D image, which are expressed as follows:

$$(X * K)_{i,j} = \sum_{m,n} X_{i+m, j+n} \cdot K_{m,n}$$

In this formula,  $X$  is the input data tensor, and  $K$  is the convolution kernel tensor.  $m$  and  $n$  indicate the height and width of the convolution kernel.  $i$  and  $j$  are the coordinate indices for the output feature map. The above equation denotes the scalar output value generated by the sliding operation when the kernel is focused at position  $(i, j)$ . This constitutes the  $(i, j)$  element of the resulting output feature map. The identical filter processes the entire image, reducing computational cost and endowing the model with translation equivariance. Another critical component of CNN is the pooling layer, compressing the spatial dimensions of the image using max or average pooling calculation. Distinguished from convolutional layers, pooling operations contain no learnable parameters and maintain the number of output channels unchanged. The convolutional architecture makes CNN ideal for image data and a cornerstone of computer vision.

However, some data distributions do not follow regular Euclidean geometry and are hardly represented in Cartesian coordinates. This is particularly true in the materials science field, where molecular structures and crystal lattices are fundamentally non-Euclidean. As a result, CNN cannot define convolution or pooling operations on these topologies and fails to accurately capture complex chemical environment information. Addressing this limitation, Graph Neural Network (GNN) provides a feasible solution [64]. Leveraging their natural ability to handle non-Euclidean data, GNNs have shown superior performance in areas like protein structure prediction, traffic prediction, and knowledge graph construction. For materials science, GNN models the material as a graph with atoms serving as nodes and chemical bonds acting as edges. Nodes encapsulate atomic features like element type and electronegativity, whereas edges bear directed or undirected connectivity details. Through the message passing mechanism,

information flows and aggregates between neighbors along edges to efficiently encode local chemical environments and global topological traits.

GNNs propagate information only along graph edges, causing a local bias that hinders direct communication between distant nodes and limits the modeling of long-range relationships. Different from CNN and GNN, the Transformer structure abandons dependence on fixed topological structures through the self-attention mechanism [65]. Data are viewed as a fully connected whole rather than discrete points. It establishes direct associations between any two nodes in a sequence or graph within a single calculation layer while remaining insensitive to their spatial or logical distance. This global receptive field is beneficial for the model to capture global information. Applications in natural language processing (NLP) and computer vision (CV) demonstrate that modeling global correlations outperforms purely local convolutions.

Transformer architecture drives the development of Large Language Model (LLM). Different from traditional DL approaches that rely on task-specific labeled data, LLM operates primarily on a self-supervised learning framework. The text data will be converted into discrete tokens and transformed into vectors through embedding. LLM directly learns the inherent structure within text, predicting the next token in a sequence based solely on the context of preceding tokens without requiring human annotation. Simultaneously, superior performance in LLM is based on massive parameter scales and substantial training costs. Taking Deepseek-R1 as a representative instance, it possesses 671 billion total parameters. However, it leverages a mixture-of-experts (MoE) architecture to activate only approximately 37 billion parameters per token during inference [66]. This sparse activation mechanism significantly enhances inference efficiency without compromising representational power. It is widely accepted that general LLM exhibits limitations within specific vertical domains. Particularly in scientific research, where they often struggle with a shallow understanding of specialized terminology, suffer from AI hallucinations. To enhance the application capabilities of general LLM in specific domains, three paradigms were proposed: prompt engineering [67], retrieval augmented generation (RAG) [68], and fine-tuning [69]. Prompt engineering is based on the principle of in-context learning. Without changing the model parameters, it guides the model's output by optimizing the input context. For example, inputting the professional knowledge into the context window enables it to organize responses in combination with the given information. However, this strategy is limited by the length of the context window and does not have long-term memory capability. In contrast, RAG expands the model's capability by constructing an external domain knowledge database. When receiving a query, the system prioritizes retrieving relevant fragments from the database and inputs them as supplementary information to the model. This effectively mitigates the AI hallucination of LLM and enhances the credibility of the results, because the model provides clear citation sources within the output. Moreover, fine-tuning is a parameterized domain adaptation strategy. By retraining on a specific domain's professional dataset, it updates some or all of the model's parameters, enabling LLMs to internalize professional knowledge and achieve deeper semantic understanding of specific domain terms and complex logic.

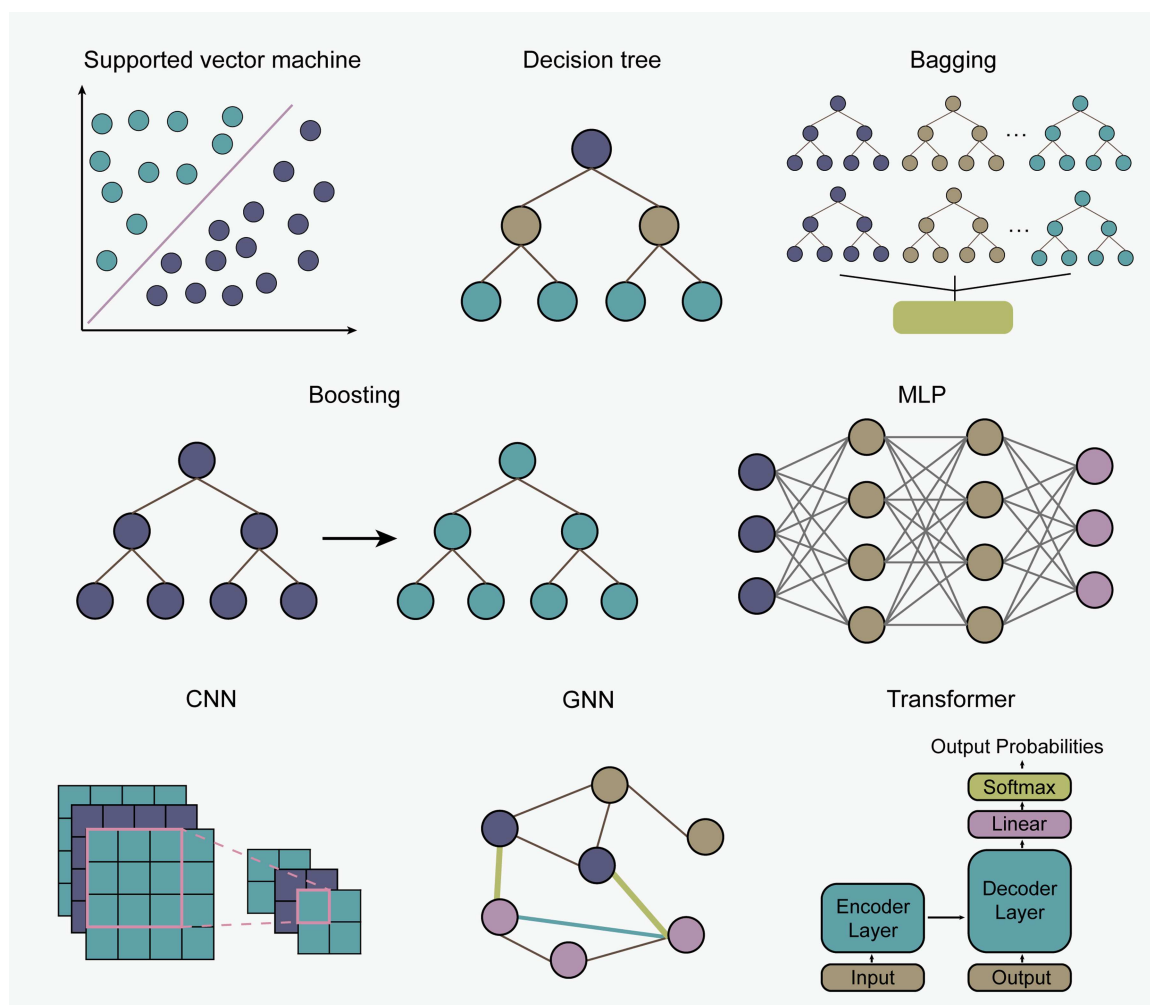
In brief, distinct ML models are suited for varying research scenarios, attributed to different core mechanisms. Linear regression and RF typically offer interpretability by delivering feature importance. Such interpretable models are beneficial for materials science research as they provide clear guidance for experimental synthesis. In contrast, most DL models are black boxes and thus less interpretable. Yet, they surpass traditional ML algorithms when processing large-scale, high-dimensional datasets. Hence, simply asserting which model is suitable for battery electrolyte research is one-sided. Researchers should evaluate computational costs, task complexity, data regime, and the requirement for reasonable choice. By flexibly selecting suitable algorithms, they can realize the deep integration of "AI for battery" (Figure 4).

## 4 | Material Representation Method

Molecular representation and crystal structure encoding are critical in bridging the gap between materials science and data science. The primary objective involves converting human-interpretable chemical formulas and material structures into machine-readable numerical tensors. The quality of material encoded directly determines training convergence and inference accuracy. Typically, material representation methods can be categorized into two paradigms: architecture-driven representation learning and knowledge-driven feature generation. Accordingly, this chapter summarizes prevailing representation methods for organic small molecules, polymers, and inorganic crystal materials. Although some methods have not been applied to the battery domain, they offer inspiring insights for future electrolyte development.

### 4.1 | Small Molecule

Molecular descriptors are typically categorized into 1D, 2D, and 3D shapes for small molecules. Simplified Molecular Input Line Entry System (SMILES) is the most widely used 1D representation. This method flattens complex molecular formulas into 1D ASCII strings [70]. For instance, ethylene carbonate is a carbonate solvent used in lithium-ion electrolytes, encoded as O=C1OCCO1. SMILES preserves atomic identities, bond connectivity, and ring information simultaneously. Moreover, reconstructing 2D Kekulé structures and 3D molecular geometries by SMILES is convenient. However, SMILES possesses intrinsic limitations as its generation depends on the choice of a starting atom and the traversal path. This leads to sequential differences of SMILES strings while corresponding to the same molecule. Recently, the development of SELFIES solved the nonuniqueness of SMILES [71]. Apart from string-based representations of SMILES and SELFIES, fixed-length binary vectors offer another efficient 1D representation. This method is usually referred to as "molecular fingerprinting," which involves transferring structural features to a bit array. These features typically include specific functional groups, atom connectivity patterns, and local topological environments. Common fingerprint schemes include one-hot encoding [72], Morgan fingerprints [73], and MACCS keys [74]. Converting varied molecular structures into vectors makes it facile for similarity assessment, enabling input to ML models. To further



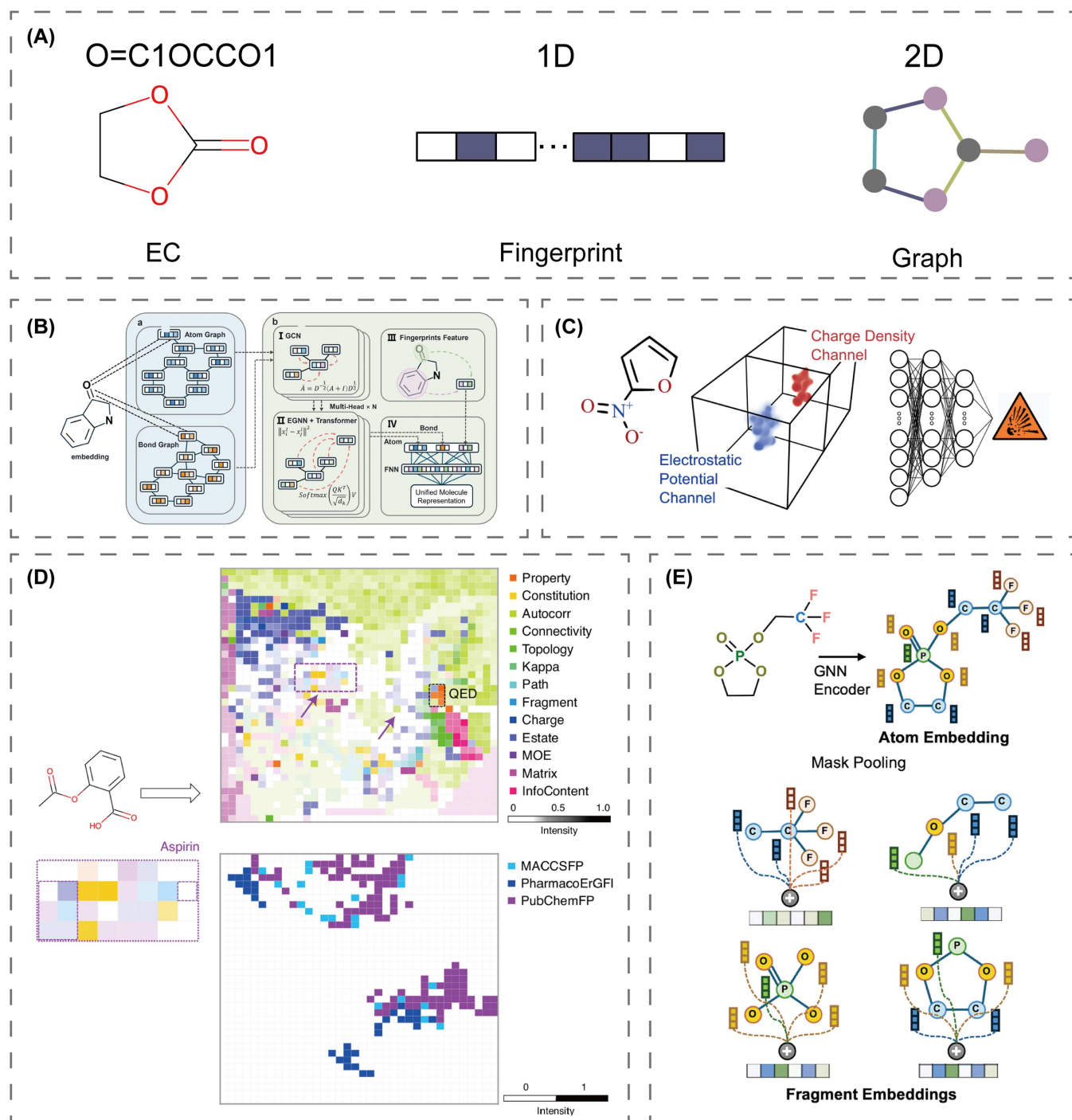
**FIGURE 4** | Schemes summary of representative ML and DL algorithms.

enrich the recognition of structural features, researchers proposed 2D molecule representation methods. In graph-based methods, GNNs are commonly employed where atoms serve as nodes and chemical bonds function as edges (Figure 5A). For instance, Qin et al. [75] proposed a multiscale feature integration model based on a GCN-Transformer architecture. This model extracts features from a molecular graph while incorporating rotational equivariance constraints and prior molecular fingerprints. Both local and global features can be extracted by this architecture. Moreover, it also introduces 3D structural information that maintains rotation and translation invariance (Figure 5B). Similarly, Zang et al. [79] developed a hierarchical molecular graph representation framework. By specifically encoding motif structures, this method extracts hierarchical representations, including the node, motif, and graph levels, respectively. Baker et al. [80] reported that CNNs can learn directly from 2D molecular images, thus eliminating the need for manually designed chemical features or fingerprints. Relying on this model, they predict key physicochemical and solubility properties, terming this approach “chemically aware.” Another significant approach is to encapsulate information into abstract feature maps using direct molecular representations. Shen et al. [77] developed a novel molecular representation method named MolMapNet. In a map, 1456 molecular descriptors and 16,204 fingerprint features of molecules are

encoded. By adopting this strategy, the model outperformed graph-based and other existing models across 26 benchmark datasets (Figure 5D).

Beyond 2D approaches, 3D molecular representations are attracting increasing attention from researchers. Electronic structures and configurations in 3D space are crucial for determining molecular properties. Based on this, Sunseri et al. [81] used multidimensional arrays of voxelised data in 3D space to model molecules. Their method supports batch sampling and data augmentation techniques, such as random rotation and translation, and can be seamlessly integrated with mainstream DL frameworks. In a distinct approach, Casey et al. [76] constructed a 3D electronic structure descriptor. They employed dual channels derived from spatial point data of electron charge density and electrostatic potential. After converting 3D information into a 4D tensor, their trained CNN model demonstrated superior performance in predicting key properties of energetic materials (Figure 5C).

Unlike fingerprints or graphs that rely solely on topological information mentioned above, using expert-generated descriptors derived from theoretical calculations provides a physics-informed approach to molecular representation. This method explicitly incorporates electronic and thermodynamic properties into the ML model, which is beneficial for guiding experiment design. Density functional theory (DFT) calculation is a



**FIGURE 5** | Classical small molecule representation method. (A) SMILES, 1D fingerprint, and 2D graph representation method of a classical solvent molecule. (B) GCN-Transformer architecture for small molecule representation [75]. Reproduced under the terms of the CC-BY license (<http://creativecommons.org/licenses/by/4.0/>). Copyright 2025, The Authors, published by Springer Nature [75]. (C) 3D representation method using electron charge density and electrostatic potential [76]. Reproduced with permission: Copyright 2020, American Chemical Society [76]. (D) 2D picture molecular representation method named MolMapNet, which encapsulates important molecule properties and fingerprints into a picture [77]. Reproduced with permission: Copyright 2021, Springer Nature [77]. (E) Scheme of the molecule is decomposed into chemical fragments and atomic structures [78]. Reproduced under the terms of the CC-BY license (<http://creativecommons.org/licenses/by/4.0/>). Copyright 2026, The Authors, published by Wiley-VCH [78].

widely used method to acquire electronic properties of solvents and additives to guide electrolyte rational design. According to frontier molecular orbital energy level theory, the energy of the lowest unoccupied molecular orbital (LUMO) determines the reduction potential of the molecule, which governs the stability

on the anode side. Meanwhile, the energy of the highest occupied molecular orbital (HOMO) is associated with the oxidation potential, dictating the high-voltage stability on the cathode side. Moreover, electronic distribution from DFT calculations describes the electrostatic environment of the molecule

properly. For instance, dipole moment ( $\mu$ ) and polarizability from DFT reflect the polarity of molecule. Moreover, the lowest negative electrostatic potential (ESPmin) ensures the nucleophilic capacity of the solvating solvent, and the weak ESPmin means decreased solvation energy. For example, He et al. [82] proposed weak ESPmin and strong highest positive electrostatic potential (ESPmax) are the main characteristics of non-solvating antisolvents. While single-molecule representation provides valuable guidelines for electrolyte design, it is significant to transfer the electrolyte entity into machine-readable language for formulation optimization. Wang et al. [78] developed a DL framework integrating intermolecular attributions across solvents with intramolecular attributions from functional units. Specifically, molecular substrate is first transformed into atom-level embeddings by a GNN encoder, which are then pooled into fragment embeddings through subgraph masking to represent functional units. After that, composition ratios, physicochemical descriptors, and salt identity are subsequently integrated via composition-aware attention to reconstruct a formulation embedding for predicting ionic conductivity. This framework achieves electrolyte entity encoding (Figure 5E). Moreover, to preserve the equivariance in mixture systems, Li et al. [83] proposed a novel geometry-aware framework named GeoMix, achieving mixed components encoding and prediction of ionic conductivity. This work extends the ideas of E(3)-equivariant neural networks (EGNN) and tensor field networks (TFN) to multicomponent mixture systems, constructing local reference frames and designing equivariant message passing mechanisms between molecules. This model is able to clearly capture the equivariance of three-dimensional spatial rotation and translation, thereby accurately modeling the geometric interactions between molecules.

## 4.2 | Polymer

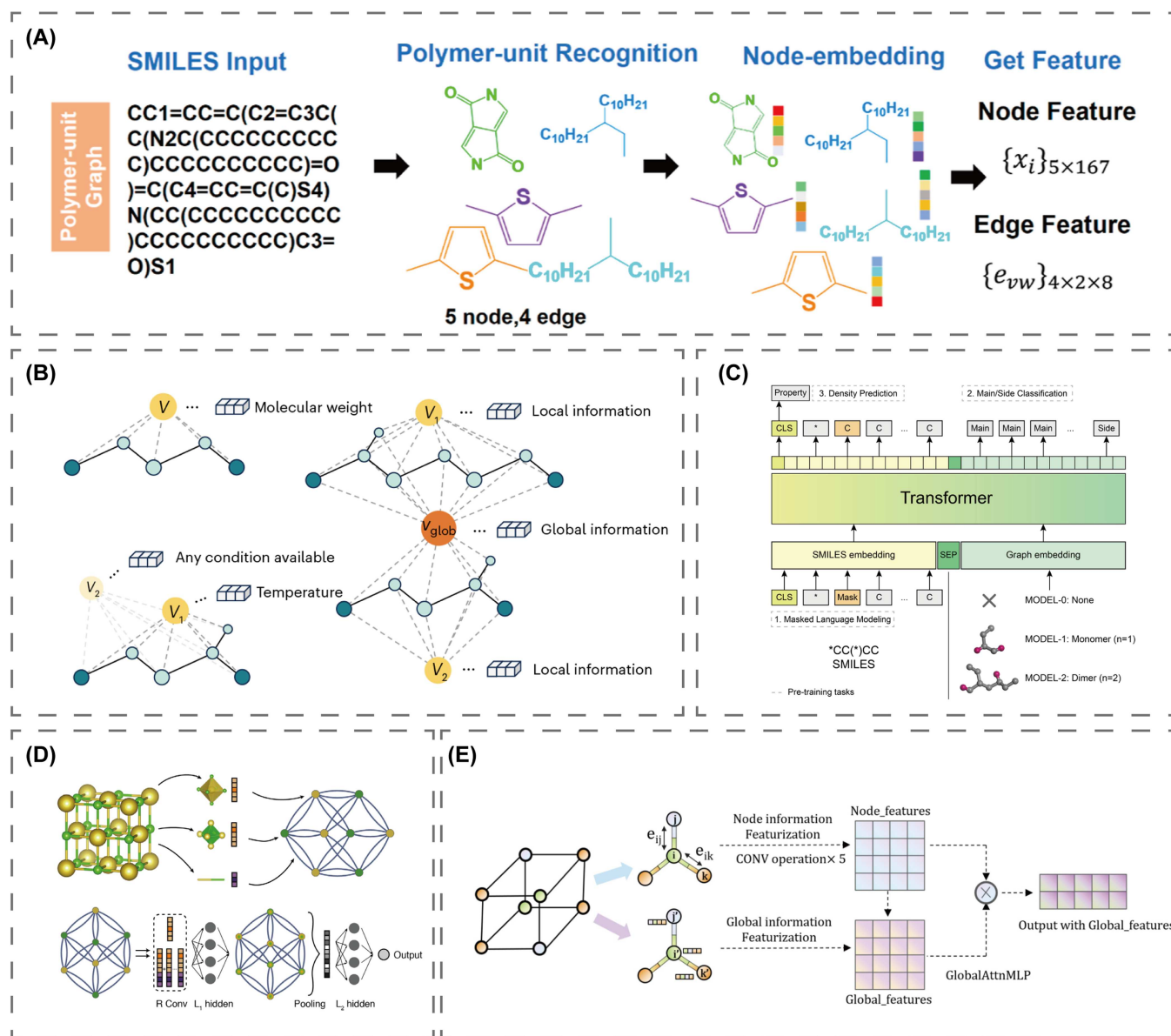
The polymer skeleton constitutes the primary component of SPEs and governs the key electrochemical performance of the battery. It is characterized by repeating subunits, variable chain lengths, and complex architectures. Critical parameters in this system include the degree of polymerization, polydispersity index, extent of crosslinking, copolymer architecture, and termination groups. Therefore, investigating polymer encoding strategies becomes essential for the ML-guided development of SPEs. A direct method is modifying traditional SMILES to accommodate polymer complexity. BigSMILES [84], Polymer SMILES (PSMILES) [85], and Group SELFIES [86] were proposed based on this method. Although linear chain structures are advantageous for sequence modeling, this approach overlooks the complex topological architecture of macromolecules. Consequently, graph-based representation methods can also be applied to describe macromolecular structures. Graph methods treat repeating units as nodes and utilize specialized edges to characterize the connectivity of macromolecules. Gurnani et al. [87] developed the PolyGNN architecture, which combines the cyclic graph of repeating unit SMILES with ML-derived atomic fingerprints. This approach preserves polymer unit invariance while accelerating feature generation, enabling accurate prediction of polymer properties. Different from traditional GNNs, a single node in the polymer unit graph represents an entire polymer repeating unit. Ye et al. [88] described polymers using a

coarse-grained approach. By integrating polymer unit graphs into GNN models, they elucidated complex structure–property relationships in organic semiconductor materials. This approach significantly simplified the molecular graph representation models. Consequently, it reduced training time by 98% compared to traditional methods (Figure 6A). Zhang et al. [89] introduced virtual nodes containing additional information of polymer, global virtual node encodes sample-level features, while local virtual nodes encode component-level features. Periodic information of the polymer was embedded in the graph rather than directly joined in a vector, which preserves the relationship and condition in polymer materials well (Figure 6B). Multimodal transformer representations demonstrate enhanced property prediction capabilities compared to single-modality models. Han et al. [90] predicted polymer properties by combining SMILES and molecular graphs. This result verifies the potential of multimodal transformers in polymer representation (Figure 6C). Xu et al. [93] proposed a transformer-based language model for property prediction named Transpolymer. In this framework, polymers were represented by sequences derived from the SMILES of repeating units and structural descriptors. Subsequently, a chemically-aware tokenizer was used to process generated sequences. With the advancement of LLM, incorporating LLM into molecular representation has emerged as an inspired direction. Zhang et al. [94] proposed a dual-modal method named PolyLLMem. This method combines text embeddings from Llama 3 with molecular embeddings from Uni-Mol. Using PolyLLMem, they successfully achieved high predictive stability for key properties such as glass transition temperature, melting temperature, and thermal decomposition temperature.

## 4.3 | Inorganic Crystal Materials

Unlike small molecules in liquid electrolytes or polymer backbones in solid polymer electrolytes, inorganic solid electrolytes are periodic crystalline materials. Their electrochemical properties are governed by elemental composition, long-range lattice order, and local coordination environments. Consequently, coding strategies for solid electrolytes draw upon representation methods used for general inorganic crystals.

Jarin et al. [95] utilized basic atomic characteristics as input features for an ML model to predict the crystal structure and lattice parameters of ABO<sub>3</sub> perovskite materials. These features consist of the atomic number, atomic mass, valence, ionic radii, electronegativity, and polarizability of the A and B sites. Finally, they achieved 88% accuracy in the classification of crystal structures and the prediction of lattice parameters. Apart from the sole atomic information, the Wyckoff position is another standard description for the symmetrically distinct atomic sites within a unit cell. Setting Wyckoff position as input feature for an ANN model, Hakami et al. [96] achieved high accuracy in the classification of six crystal structures, including perovskite, layered perovskite, spinel, fluorite, halite, and ilmenite. Periodic crystal structures can also be described as a data format based on graphs. Similar to small molecules and polymers, nodes represent atoms and edges represent chemical bonds. Xie et al. [91] proposed a graph convolutional neural network (GCN) framework in their study to predict key characteristics of crystal structures. They first transformed the crystal structure into graphs. After constructing the crystal graph, this structure was



**FIGURE 6** | Polymer and inorganic materials representation method. (A) Polymer representation based on GNN [88]. Reproduced with permission: Copyright 2024, American Chemical Society [88]. (B) The introduction of virtual nodes on polymer graph representation [89]. Reproduced with permission: Copyright 2024, American Chemical Society [89]. (C) Multimodal transformer combined SMILES and molecule graph for polymer segments representation [90]. Reproduced with permission: Copyright 2024, American Chemical Society [90]. (D) Workflow of crystal lattice representation method based on GCN [91]. Reproduced under the terms of the CC-BY license (<http://creativecommons.org/licenses/by/4.0/>). Copyright 2021, The Authors, published by Springer Nature [91]. (E) Architecture of Gformer for crystal representation [92]. Reproduced with permission: Copyright 2025, Elsevier [92].

input into a CNN. Following convolution and pooling operations, the output layer delivers the final prediction results. This framework reliably estimates important properties calculated via density functional theory based on the structure and composition of inorganic crystals (Figure 6D). Isayev et al. [97] reported a descriptor method based on GNNs, named property-labeled materials fragments (PLMF). They evaluate potential bond distances and the topological structure of neighboring atoms to capture details of local structures. Ultimately, this approach successfully achieves accurate predictions for key properties, including metal/insulator classification, bandgap energy, bulk modulus/shear modulus, Debye temperature, and heat capacity. To further enhance the capability in capturing

global information and periodic patterns within crystal structures, Wang et al. [92] proposed a novel model named Gformer. It consists of periodic pattern encoding, self-connecting edges, graph attention convolutions, and global feature extraction techniques. Within this model, information from adjacent nodes is integrated through a message-passing mechanism. Simultaneously, the graph attention mechanism refines the extraction of node and edge features, which is beneficial for learning complex chemical bonding patterns. For global feature extraction, the model analyzes elemental distributions within the dataset, introducing global elemental composition features to reflect the overall chemical information of materials. Ultimately, outstanding performance was achieved, including the

prediction of bandgap, formation energy, and shear modulus relying on Gformer (Figure 6E). In addition, voxel images of crystals can be input into CNNs. Davariashtiyani et al. [98] utilized normalized atomic numbers, periods, and groups from the periodic table as RGB channels to color voxels occupied by atoms, constructing a pure visual representation without physicochemical properties. Additionally, to enhance the model's robustness against crystal rotational invariance, they introduced random rotations during model training. Ultimately, this verified the superior capability in property prediction. However, precise 3D crystal coordinates are scarce during early-stage high-throughput screening. To bridge this gap, Li et al. [99] developed a stoichiometry-based representation learning named MEMFNet, which does not require prior structural information. Based on the Roost GNN architecture, MEMFNet virtualizes chemical formulas into physically constrained graphs, thereby mimicking the octahedral framework of layered materials without specific coordination information. Meanwhile, advancements in language models have made converting crystal structures into textual data increasingly vital. Kim et al. [100] transformed CIF-formatted structural data from the Materials Project into textual descriptions, which are suitable for LLM input prompts by utilizing Robocrystallographer [101]. Employing stoichiometric formulas and textual structure descriptions, they fine-tuned GPT-4o-mini for the general synthesizability prediction task. The study concluded that LLM-embedding vectors serve as an effective structural representation for complex structures, offering a viable alternative to conventional graph-based formulations.

## 5 | Application of ML in Electrolyte Development

After establishing the fundamental mechanisms of electrolytes and the introduction of ML methodologies, we have discussed representative molecular representations that link data science and materials science. In this section, we will present detailed case studies of ML-driven electrolyte development, focusing on liquid electrolytes, SPE, and ISE.

### 5.1 | ML for Liquid Electrolyte

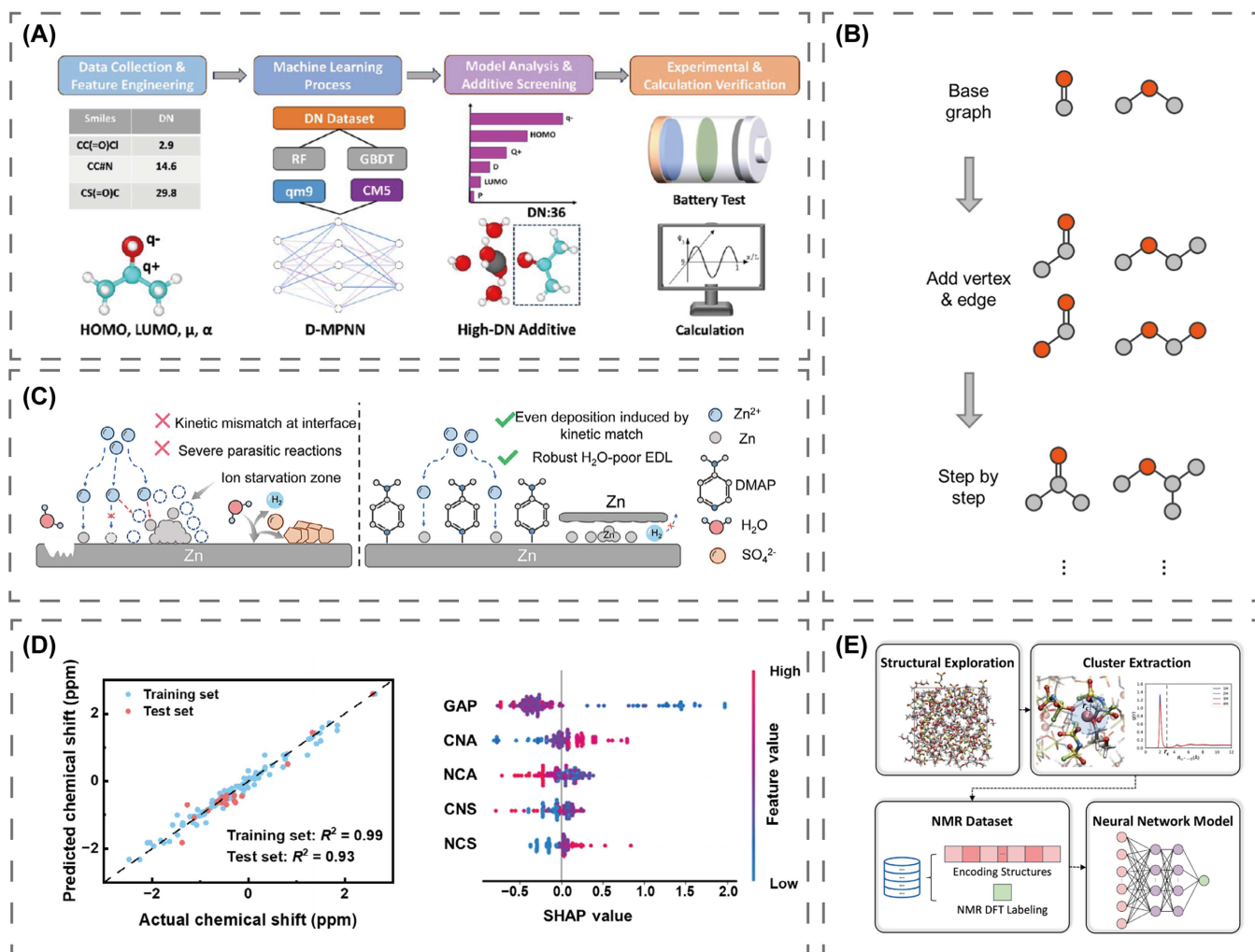
Research on ML for liquid electrolytes can generally be categorized into two main areas. The first focuses on using DL, such as ANN, to predict physicochemical properties, covering both intrinsic physicochemical properties and macroscopic electrolyte properties (e.g., ionic conductivity, viscosity). Moreover, ML-guided materials discovery has significantly accelerated the exploration of suitable candidates for battery applications. This approach largely consists of two further strategies: knowledge discovery for extracting molecular design rules, and high-throughput virtual screening (HTVS) for molecule discovery.

#### 5.1.1 | Prediction of Physicochemical Properties

The preparation of liquid electrolytes encompasses the selection, design, and synthesis of high-performance solvent and salt components. As for the solvent component, several critical properties exert a decisive influence on the overall electrolyte performance. Melting point (MP), boiling point (BP), and flash

point (FP) of electrolyte molecules are essential for battery development. Gao et al. [102] reported a knowledge-data dual-driven framework named KPI for properties prediction. The KPI achieved low mean absolute errors of 10.4, 4.6, and 4.8 K for MP, BP, and FP predictions, serving as an advanced tool for electrolyte development. Furthermore, Guo et al. [103] proposed that electrolytes with low MP, high BP, and high dielectric constant show outstanding performance under low-temperature conditions. Hence, they integrated LLM, model construction, and interpretability, predicting key molecule properties. This work confirmed that molecule size and hydrogen-bonding capacity influence BP, while intermolecular interactions determine MP. Qin et al. [104] developed an intelligent screening workflow encompassing self-supervised learning, BP prediction, solvation calculations, and experimental validation. They revealed that the nitrile group broadens the liquid range and significantly enhances dielectric constants, whereas ether bonds adversely affect boiling and melting points. Based on these findings, 3-methoxypropionitrile was ultimately selected as the primary electrolyte solvent, enabling stable battery operation within the  $-60^{\circ}\text{C}$  to  $120^{\circ}\text{C}$  range. Moreover, Gutmann's donor number (DN) and acceptor number (AN) values serve as pivotal indicators. These metrics characterize the polarity, solubility, and compatibility of solvent molecules. Many research studies have established electrolyte design principles based on DN values [105, 106]. However, accurately obtaining these two metrics is both costly and time-consuming. Therefore, acquiring these values by an ML model is meaningful. Liu et al. [107] investigated four ML models for predicting DN and AN. Among these, the CatBoost model achieved test  $R^2$  exceeding 0.86 and 0.96 for the prediction of DN and AN. This approach reduces the need for costly experiments and theoretical calculations. Furthermore, through interpretable analysis, they revealed that features TDB02m and HATS1s exert the greatest influence on DN and AN. Luo et al. [108] proposed an architecture including ML, DL, theoretical calculations, and experimental verification. This framework aims to efficiently screen additives for aqueous zinc-ion batteries. Specifically, they first explored the intrinsic relationship between DN and selected relevant descriptors based on a collected dataset of organic molecules. Furthermore, Shapley Additive Explanations (SHAP) analysis was used to demonstrate feature importance. After that, guided by this DN prediction model, isopropanol was identified as an ideal additive with a high DN value. When introduced into aqueous electrolytes for model validation, the battery maintained up to 99% coulombic efficiency (CE) after 450 cycles. (Figure 7A). Furthermore, the dielectric constant is another crucial indicator of solvent polarity. Yao et al. [113] employed an ANN to predict the dielectric constants of mixed systems containing ethylene carbonate and salts. Their predictions showed high consistency with molecular dynamics (MD) simulation results. After that, they integrate findings with experimental characterization, revealing the atomic-level chemical origins of dielectric constant variations. This work provides valuable insights into the solvation regulation of electrolytes.

According to the molecular orbital theory, electrochemical behavior relates to the frontier orbitals of molecules. HOMO is primarily associated with oxidation tendencies, while LUMO is typically used to assess reduction tendencies. Although DFT calculations can get the HOMO and LUMO, it is



**FIGURE 7** | Properties prediction related to liquid electrolytes (A) Donor number prediction model workflow of solvent molecule [108]. Reproduced with permission: Copyright 2025, American Chemical Society [108]. (B) Graph-based molecule generation process for LUMO prediction [109]. Reproduced with permission: Copyright 2023, American Chemical Society [109]. (C) Zn battery additive decomposition mechanism on the surface of Zn metal [110]. Reproduced with permission: Copyright 2025, Wiley-VCH [110]. (D) NMR prediction result of <sup>7</sup>Li and the corresponding SHAP analysis [111]. Reproduced with permission: Copyright 2025, Wiley-VCH [111]. (E) Framework of NMR spectra prediction model [112]. Reproduced with permission: Copyright 2025, American Chemical Society [112].

computationally expensive in high-throughput conditions. Therefore, replacing DFT calculations with ML to predict HOMO and LUMO levels proves highly attractive for data-driven electrolyte development. Gao et al. [109] employed a graph-theoretic approach to establish a dataset comprising 1399 solvent molecules. This study revealed that after these molecules interact with Li<sup>+</sup>, increasing Li–O binding energy, bond length, and C–O bond length will decrease their LUMO. Further interpretable analysis indicated that the 2p orbitals of carbon atoms significantly contributed to the LUMO energy level changes. Guided by these, researchers achieved LUMO prediction based on an RF model (Figure 7B). Han et al. [110] discovered that HOMO is a key descriptor for predicting the surface adsorption energy between metallic Zn and a molecule. Based on this, they constructed a four-descriptor model to predict the HOMO values of molecules. This study confirmed that polarizability and atomic charge are the most critical descriptors for determining the HOMO energy level, a

conclusion strongly supported by experimental evidence. They employed 4-dimethylaminopyridine as an additive to suppress zinc dendrite growth and further enhance the credibility of ML results. Its mechanism of action involves slowing the rate of interfacial charge transfer and alleviating the local ion depletion state (Figure 7C).

Moreover, viscosity represents another critical evaluation metric for electrolytes, as high viscosity hinders ionic migration, thereby reducing ionic conductivity. Furthermore, high-viscosity electrolytes struggle to permeate the micropores of electrode materials. These issues hardly form stable electrolyte–electrode interphases. Therefore, employing ML for viscosity prediction is beneficial for rapid solvent screening and composition optimization. Azeez et al. [114] reported a hybrid ML framework combining a single-parameter semi-empirical model with an ANN. This framework aimed to predict the viscosity of salt-free solvent mixtures and the relative viscosity of lithium-ion electrolyte solutions. Seven parameters were selected as inputs for the artificial neural

network. These parameters included salt concentration, electrolyte temperature, salt anion size, solvent melting and boiling points, solvent dielectric constant, and solvent dipole moment. Results demonstrated the superiority of their framework. For binary mixtures, the overall mean absolute percentage error (MAPE) was 2.05%, and for ternary mixtures, it was 3.17%. Additionally, the MAPE for predicting the relative viscosity of electrolyte solutions using the artificial neural network model was 4.86%. The ESW of the electrolyte determines the energy density of the battery device directly by influencing the work voltage. A high oxidation potential (OP) delivers a wide ESW, which corresponds to enhanced energy density. Hence, utilizing data-driven methods to predict the OP is a valuable direction. Xu et al. [115] established an ML framework to predict the OP of organic molecules in the electrolytes of ZIBs. The ESW of 307 organic solvents was investigated systematically. Different ML models, including SVM regression, kernel ridge regression, XGBoost, and Gaussian process regression, were used to enable the facile prediction of OP. Moreover, relying on the SHAP values calculation, the key features that determine OP were identified (N3, F1, O1, C3). Finally, the reliability of these ML results was verified in Zn symmetric cell tests, which operated stably for more than 500 h.

Li nuclear magnetic resonance (NMR) is a powerful way to investigate the solvation structures by distinguishing the  $\text{Li}^+$  chemical shift. Moreover, distinct from covalent and ionic bonds, the Li bond is recognized as an analog of the hydrogen bond [116]. It reflects the interaction between  $\text{Li}^+$  and other items in battery systems. Recently, Gao et al. [111] constructed an ensemble ML model combining gradient boosting regression and SVM regression. Utilizing the anion-solvent coordination number and the component-weighted average HOMO–LUMO gap as key descriptors, they achieved a credible prediction of  $^7\text{Li}$  NMR chemical shifts. The test set coefficient of determination  $R^2$  reached 0.93. The SHAP analysis revealed that the HOMO–LUMO gap is the key feature determining the  $^7\text{Li}$  NMR results. Furthermore, it verified that the anion influences the NMR results more significantly than the solvent, which accelerates the advancement of the Li bond theory (Figure 7D). In addition, linking the relationship between solvation structures and  $^7\text{Li}$  NMR spectra presents a significant challenge. Traditional DFT calculations hardly provide dynamic local information as they merely yield static solvated structures. The experimental chemical shift values originate from a weighted average of different local positions, incorporating detailed information about local structure and dynamics overall. To resolve this issue, You et al. [112] proposed a novel framework comprising two ML models. An MLP model was utilized to generate trajectories across a range of concentrations using DFT-labeled data, while a separate ANN model was trained to predict the NMR spectra. Moreover, they found that abnormal reversal occurs as the salt concentration approaches its limitation. This phenomenon observed in experimental characterization was successfully reproduced in the ML predicted results, demonstrating the robustness of this workflow (Figure 7E).

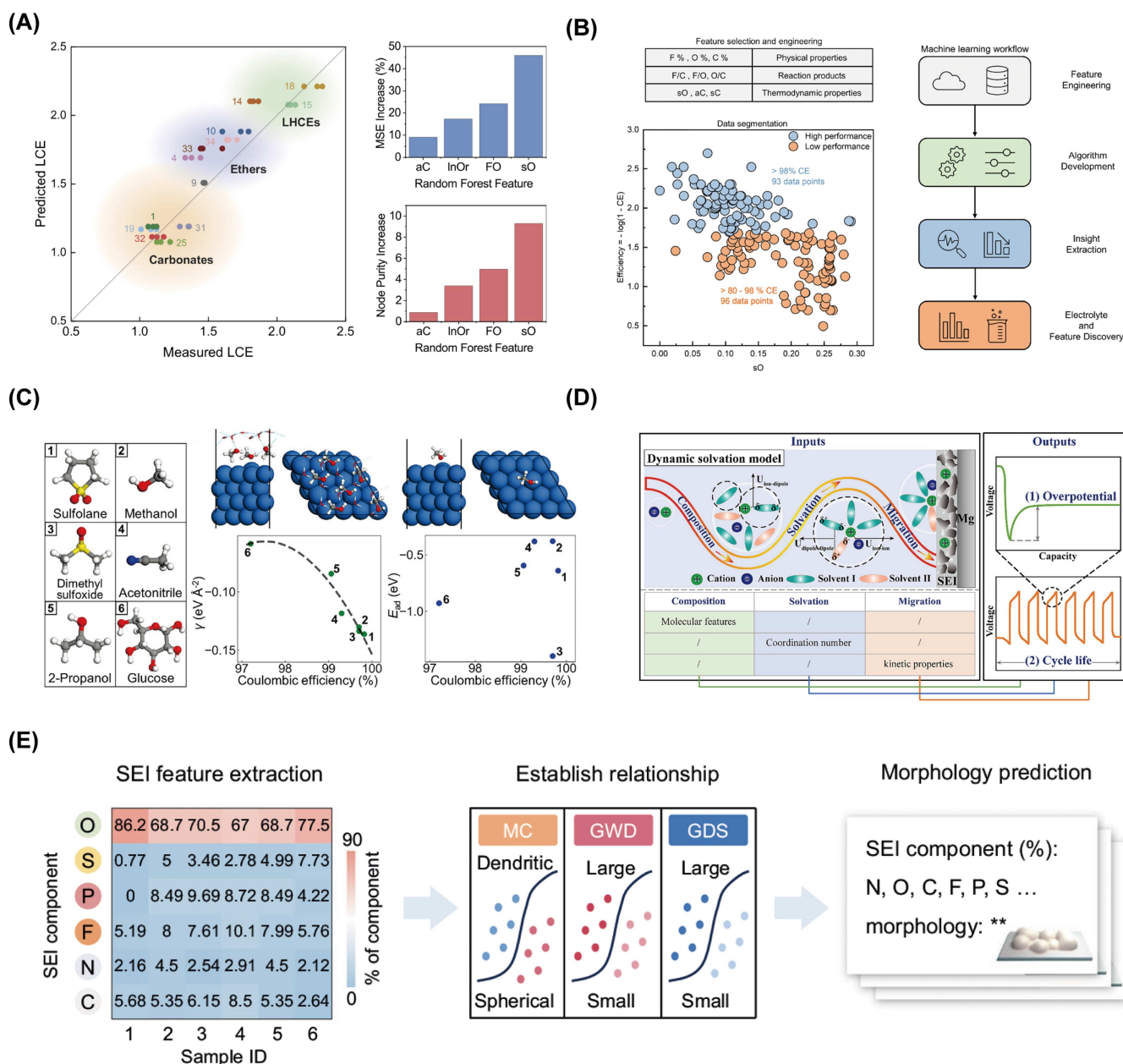
### 5.1.2 | ML-Guided Molecule Exploration

Complex and nonlinear relationships of materials are difficult to decipher, relying on statistical methods and human intuition.

Beyond property prediction, ML-guided molecule exploration based on large-scale data attracts significant attention. This capability is particularly suitable for the liquid electrolyte domain, where the space of molecular structures is infinite. Consequently, research directions in ML-guided molecular design can be categorized into two primary streams, comprising knowledge discovery based on interpretable models and virtual screening based on black-box models [117].

CE of the Li–Cu cell is recognized as an important index reflecting the properties of the electrolyte/Li interphase, which is a significant indicator for liquid electrolyte [118]. Kim et al. [119] collected CE data for Li–Cu cells from reported literature, establishing a dataset containing 150 samples. They designed the logarithmic CE (LCE) as the target label and extracted element ratios as input features. Following hybrid feature selection, the linear regression model revealed that a lower solvent oxygen content (sO) correlates with higher LCE. Guided by this, this team investigated solvents including methyl butyl ether (MBE), methyl t-butyl ether (MTBE), and dibutyl ether (DBE) with different sO values of 0.056, 0.056, and 0.037, respectively. Consistent with the model prediction, DBE with the lowest sO exhibited the highest CE up to 99.7% (Figure 8A). Subsequently, Oyakhire et al. [120] divided the electrolyte dataset into different CE ranges and constructed distinct linear regression formulas for each. They concluded that lower sO is beneficial for both ranges of electrolytes, while higher sC only influences high-CE samples (Figure 8B). Li et al. [121] utilized SISSO [124] based on linear regression to uncover the relationship between CE and the surface free energy ( $\gamma$ ) of the Zn (002) surface. This model concluded that a lower number of heavy atoms (#HA) promotes tighter molecular packing, while increasing surface tension ( $\sigma$ ) enhances intermolecular interactions. Both factors promote the formation of a stable Helmholtz layer on the electrode surface, thereby enhancing the stability of aqueous zinc-ion batteries. Based on these insights, they screened additives and identified seven molecules with the lowest  $\gamma$  values. Among these, acetone and butanetriol exhibited the most outstanding performance in terms of CE, voltage polarization, and long-term cycling stability (Figure 8C).

The aforementioned models are primarily based on linear frameworks, which offer inherent interpretability but perform poorly in handling complex nonlinear relationships. Unlike models with inherent interpretability, most ML architectures rely on post-hoc interpretability methods such as SHAP and locally interpretable model-agnostic explanations (LIME). To correlate micro-solvation structures with macro-electrochemical performance, Li et al. [122] proposed a dynamic solvation model that concentrated on three distinct electrolyte levels, including molecular characteristics, coordination number, and kinetic properties. Relying on feature engineering and SHAP analysis, they discovered that increasing the content of amino ( $\text{NH}_2$ ) groups while decreasing carbonyl ( $\text{C=O}$ ) groups of the molecule reduces deposition overpotential. It is widely accepted that a lower LUMO corresponds to anion-drive SEI. At the solvation level, this work found that reducing LUMO by maintaining coordination numbers below 5 and weakening the Mg–O bond strength is practical. At the kinetic properties level, they found that adjusting the dielectric constant and carbon chain length is significant. Based on insights related to the three levels, the optimally selected electrolyte NETEN + TFE/Mg(OTf) $_2$  exhibits remarkable electrochemical performance (Figure 8D). Chang et al. [125] combined domain knowledge with data-driven



**FIGURE 8** | Knowledge discovery of liquid electrolytes. (A) CE prediction by linear model and range of feature importance [119]. Reproduced under the terms of the CC-BY-NC-ND license (Deed—Attribution-NonCommercial-NoDerivatives 4.0 International—Creative Commons). Copyright 2023, The Authors, published by PNAS [119]. (B) Prediction for different electrolytes at different CE ranges [120]. Reproduced with permission: Copyright 2025, Royal Society of Chemistry [120]. (C) High-performance additive discovery for Zn battery [121]. Reproduced with permission: Copyright 2024, Wiley-VCH [121]. (D) Dynamic solvation model for Mg battery development, including composition, solvation, and migration level [122]. Reproduced with permission: Copyright 2025, Royal Society of Chemistry [122]. (E) ML workflow for SEI element and Lithium deposition morphology investigation [123]. Reproduced with permission: Copyright 2025, Springer Nature [123].

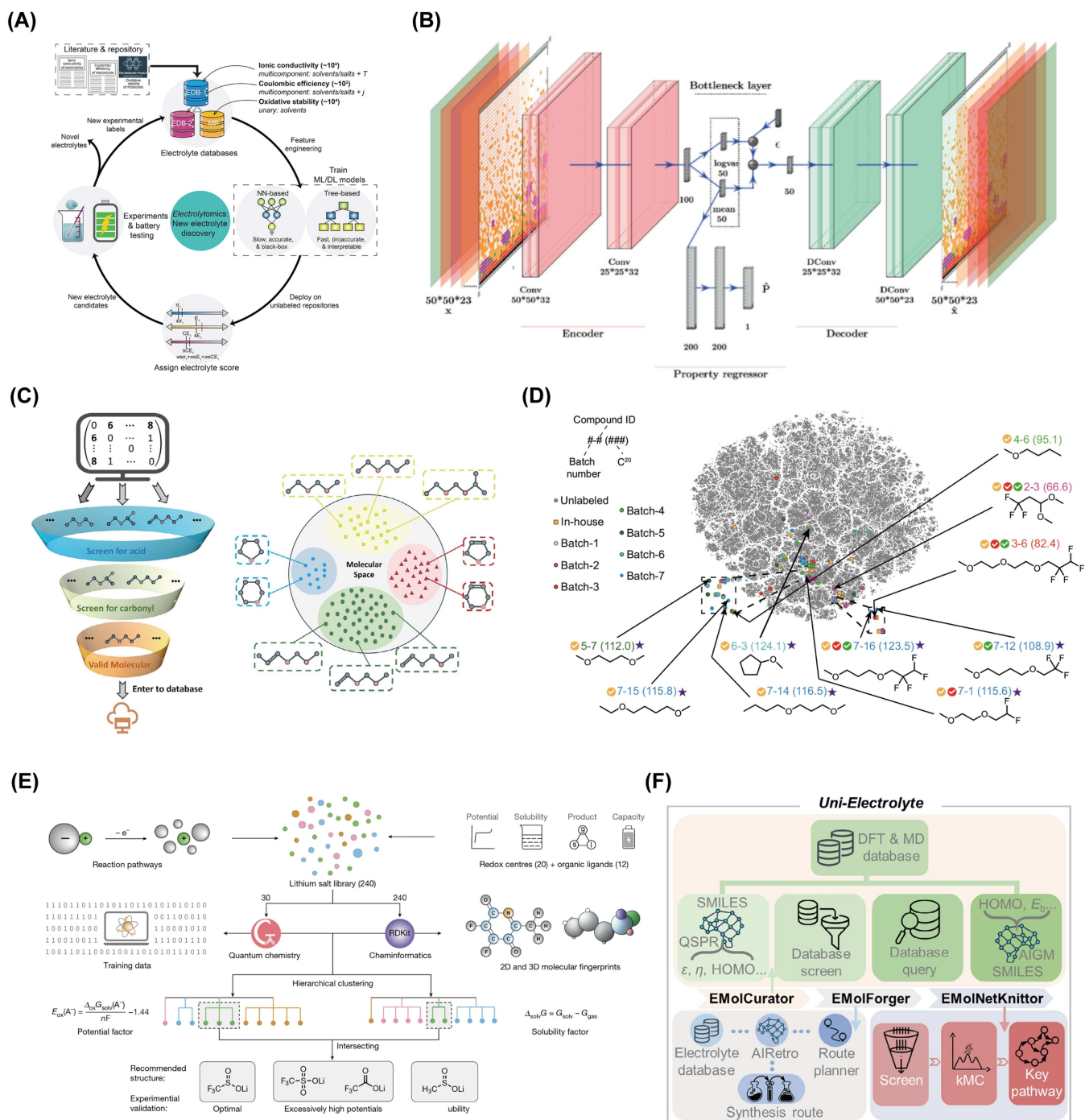
methods to establish a design workflow for zinc-ion batteries, encompassing data collection, feature engineering, and model evaluation. This workflow identified five key factors influencing ionic conductivity, including temperature,  $\text{Zn}^{2+}$  molar ratio,  $\text{DN}_{\text{anion}}$ ,  $\text{DN}_{\text{solvent}}$ , and  $\mu\text{H}_2\text{O}$ . Guided by SHAP values, they rationally designed the  $\text{Zn}(\text{ClO}_4)_2\text{-TMU}$  electrolyte that exhibited high ionic conductivity and stable cycling performance in zinc symmetric cells. Wang et al. [126] developed a physics-information-driven ensemble learning framework based on a

MoE. This framework considers four dimensions of molecules: chemical composition, elemental ratios, physicochemical properties, and topological structure. Consequently, this MoE framework showed higher robustness than traditional ML models in surface free energy ( $\gamma$ ) prediction. By feature engineering and SHAP analysis, this work discovered that a higher N/O ratio contributes to optimized surface stability. Based on this insight, disodium ethylenediaminetetraacetate ( $\text{Na}_2\text{EDTA}$ ) and ethylenediaminetetraacetic acid (EGTA) were identified as high-performance

additives for zinc-iodine flow batteries. Besides, designing electrolytes from the interfacial perspective is interesting. Lu et al. [123] developed a hybrid framework incorporating Cryo-transmission electron microscopy (Cryo-TEM) analysis, ML, and multiscale theoretical simulations. This framework investigated the relationship between the chemical composition of SEI and the morphology of lithium metal deposition. Relying on an ensemble model, they conducted three ML tasks, including morphology classification, diameter of spherical Li prediction, and width of dendritic Li prediction. Finally, SHAP analysis discovered that elevated N, S, P, and F contents alongside reduced O ratios in the SEI are favorable for Li deposition. Guided by these, an O-free additive named HFTP was added to the electrolyte. This strategy achieved a highly disordered SEI and uniform Li deposition, enabling stable operation for 800 cycles in Li-Cu cells and 220 cycles in 1 Ah Li||NCM811 pouch cells (Figure 8E).

In contrast to knowledge discovery, black-box models combined with virtual screening are employed to filter target candidates from large-scale datasets. Zhang et al. [127] established a data-knowledge dual-driven approach to screen the solvent molecule for a high-temperature fast-charging electrolyte. This framework consists of high-throughput calculations, ML, and experimental verifications. They effectively screened ethyl trimethylacetate solvent and enabled fast-charging (15-min charging to 80% capacity) cycling without Li plating over 4100 cycles. Kumar et al. [128] proposed the concept of “electrolytomics” by establishing a comprehensive workflow to develop novel electrolytes for lithium metal batteries and high-voltage lithium ion batteries. Utilizing LightGBM and Chemprop ML models, this workflow predicts ionic conductivity, oxidation stability, and CE simultaneously. Notably, it achieves results similar to those of molecular dynamics simulations without the explicit encoding of ionic solvation. Integrating these three metrics, the authors proposed an electrolyte score (eScore) to screen a molecular database containing approximately 185 million entries. Finally, an ethanesulfonyl fluoride-based electrolyte was identified and verified, enabling high ionic conductivity, high-voltage stability, and wide-temperature feasibility (Figure 9A). Deep understanding of the SEI is pivotal for electrolyte development. Rajagopal et al. [129] proposed a materials analysis and synthesis framework for SEI-related virtual screening, leveraging Kinetic Monte Carlo (KMC) simulations. Firstly, they generated 50,000 two-dimensional SEI structures via KMC. And they employed a variational auto-encoder (VAE) model combined with a property predictor to investigate SEI structures. Through comprehensive analysis of SEI thickness, density, porosity, and volume fraction, this study provides valuable reverse-engineering insights into the structural and reaction barrier space of SEI (Figure 9B). In the field of magnesium battery electrolyte development, Gao et al. [130] first used a molecule generation method with prior knowledge to construct a dataset of 823 virtual organic molecules. Subsequently, they used QM calculations to investigate physico-chemical properties of the generated molecules, revealing that electronic properties are primarily determined by bond type rather than molecular topological structure. After systematically investigating the related electronic properties, the authors established detailed selection guidelines. They filtered 18 potential candidates from the virtual dataset and compared them with common high-performance ether-based solvents

(e.g., DME, DOL). Finally, screened dioxane exhibited superior anodic stability to DME, maintaining a CE of up to 99.54% for 5200 cycles, which verified the reasonability of the ML workflow (Figure 9C). In the field of aqueous Zinc-ion batteries (AZIBs), employing organic molecule additives could stabilize the Zinc anode. Xu et al. [134] considered three key properties, including molecule adsorption energies on the surface of Zn metal, redox potentials, and water solubility. Finally, they screened 75,024 organic molecules and identified cyanoacetamide (CA) and hydantoin (HN) as novel additives. This work exemplifies the scalability of AI-driven screening in aqueous chemistries. Active Learning (AL) demonstrates significant potential in scarce data scenarios. By selecting the most valuable samples for annotation, AL substantially reduces data annotation costs. This character makes it suitable for materials-related tasks. Combining AL with virtual screening has shown exceptional efficiency in electrolyte development. Ma et al. [131] constructed a dataset recording the capacity retention of Cull-LiFePO<sub>4</sub> to train a Gaussian Process Regression (GPR) surrogate model. After that, they established a screening space comprising 135 million molecules downloaded from eMolecules and PubChem. After removing redundant chemical groups, conducting synthetic feasibility analysis, and applying conductivity screening, 3.8 million unique solvent molecules were selected. After that, they employed a trained GPR surrogate model to predict and rank samples within this virtual chemical space, followed by experimental performance testing of top-ranked candidate materials. Through cycles of feeding experimental data back into the training process and iterative refinement, the model gradually converged and identified the optimal electrolyte candidates. Ultimately, they conducted comprehensive investigations into the solvation structures, SEI positions, and lithium deposition morphologies of the four optimal candidate materials. Results revealed that the electrolytes discovered by AL exhibited similar interfacial characteristics (Figure 9D). Moreover, to address the challenges posed by unlabeled datasets, unsupervised learning algorithms offer powerful processing capabilities for molecular discovery. In battery manufacturing, Li<sup>+</sup>/Na<sup>+</sup> supplement additives are significant, concerning electrochemical activity, redox potential, decomposition products, and solubility. To explore this, Chen et al. [132] generated a Lithium salt library of 240 samples by combining 20 redox centers and 12 organic ligands. Following the establishment of potential and solubility descriptors, the team performed sample screening using hierarchical clustering. Identifying LiSO<sub>2</sub>CF<sub>3</sub> as the best candidate for external Li supply, the study highlighted its impressive compatibility in configurations such as anode-less Cr<sub>8</sub>O<sub>21</sub>, sulfurized polyacrylonitrile, and graphite/LiFePO<sub>4</sub> cells. This approach provides a brand-new life-prolonging method for battery technology (Figure 9E). Additionally, this unsupervised learning strategy was further validated in the domain of presodiation additive discovery [135]. A complete AI platform accelerates molecule discovery effectively. Chen et al. [133] constructed a Uni-Electrolyte platform that contains three parts for molecule discovery. Firstly, EMolCurator leverages quantitative structure-property relationship (QSPR) models, multi-criterion screening, similarity search, and molecular generation to explore promising candidates. After that, EMolForger provides optimized synthetic routes and detailed reagent information. Finally, EMolNetKnitter investigates crucial battery interphase



**FIGURE 9** | Virtual screening for electrolyte components. (A) Electrolytics workflow for electrolytes development [128]. Reproduced with permission: Copyright 2025, American Chemical Society [128]. (B) CNN framework for SEI-related virtual selection [129]. Reproduced with permission: Copyright 2023, Wiley-VCH [129]. (C) HTVS for the electrolytes of Mg battery electrolyte [130]. Reproduced with permission: Copyright 2025, Wiley-VCH [130]. (D) HTVS combined AL for anode-free lithium metal battery electrolytes [131]. Reproduced with permission: Copyright 2025, Springer Nature [131]. (E) Unsupervised learning for lithium additive exploration and the corresponding decomposition mechanism [132]. Reproduced with permission: Copyright 2025, Springer Nature [132]. (F) Workflow of Uni-Electrolyte, an AI-driven electrolyte development platform [133]. Reproduced with permission: Copyright 2025, Wiley-VCH [133].

insights, providing SEI composition and formation mechanisms (Figure 9F). However, the cases mentioned above mainly focused on the solvent and salt, respectively. The electrochemical performance of liquid electrolytes was determined by multiple mixtures of solvents, salts, and additives. So, establishing a

robust method to explore multiple components of electrolyte formulation from an enormous design space is meaningful. Yang et al. [136] proposed a physics-informed framework that preserves permutation invariance and incorporates dependencies on temperature and salt concentration, making it

applicable to property prediction tasks of molecular mixtures. The forward prediction part of this framework adopts a three-stage process. Firstly, through graph neural networks, the general molecular embeddings are pretrained on over 240,000 single-molecule data. Then, using permutation invariant neural networks, the molecular embeddings of each component are aggregated into the overall electrolyte embedding according to learnable molar ratio weights. This aggregation model is pretrained based on over 100,000 molecular dynamics simulation data. Finally, combined with over 10,000 experimental conductivity data and physical priors, it is fine-tuned to accurately predict ionic conductivity and anion coordination ratios. Relying on the novel embedding method of molecular mixtures, screening multiple-component electrolyte formulations becomes possible.

## 5.2 | ML for Solid-State Polymer Electrolytes

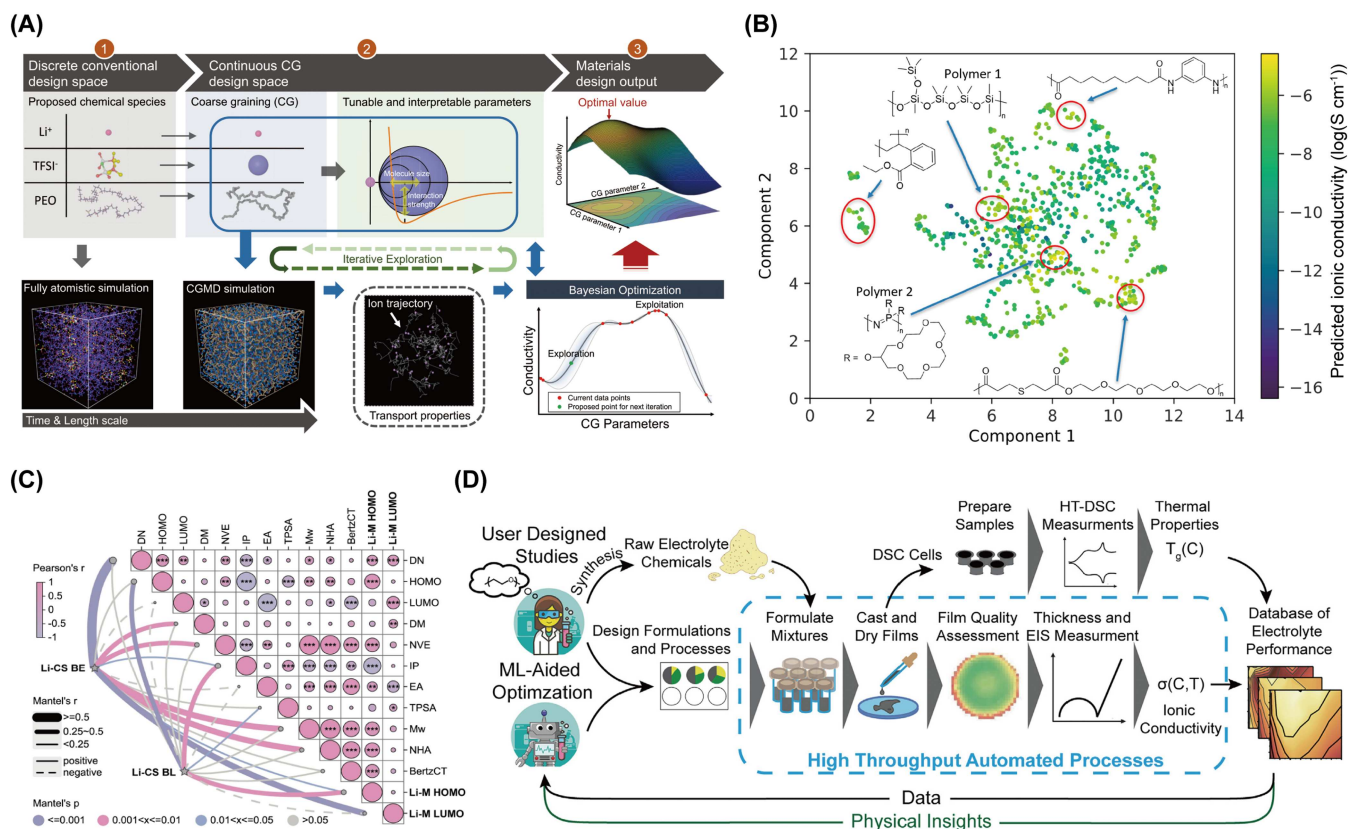
SPE has become an important research direction for the next generation of battery technology because of its potential to improve safety, energy density, and prolong life in secondary ion batteries. However, low ionic conductivity at room temperature limits its practical application. ML provides a new way to overcome this challenge and accelerate the development of high-performance SPE.

ML guided the development of SPE with high ionic conductivity by establishing the relationship between molecular structure and ionic conductivity [137]. The microstructure, molecular composition, and interaction of polymer electrolytes are usually modeled by molecular dynamics and coarse-grained molecular dynamics (CGMD) simulations. Wang et al. [138] developed a new SPE design method that combines CGMD with ML. They constructed a continuous high-dimensional design space composed of physically interpretable general descriptors by coarse-grained processing of chemical substances. Then, the Bayesian Optimization algorithm was used to efficiently explore the space through autonomous CGMD simulation. This method obtains a comprehensive description of the relationship between lithium conductivity and intrinsic material properties, such as molecular size and non-bond interaction strength at the molecular level (Figure 10A). This work provides guidance for optimizing the composition of SPE, including anions, secondary sites, and main chain structure, accelerating the development of highly conductive SPE. Moreover, Bradford et al. [139] developed a chemical information ML model that can accurately predict the ionic conductivity of SPE. The model encodes the Arrhenius equation describing the temperature activation process into the readout layer of the message passing neural network. Compared with the uncoded temperature-dependent model, the prediction accuracy is significantly improved. The model predicts the ionic conductivity of thousands of SPE candidates to identify promising candidate materials, and predicts different anions in polyethylene oxide and poly(trimethylene carbonate), which proves the practicability of the model in identifying SPE ionic conductivity descriptors (Figure 10B).

In addition, ML discovers novel polymer electrolytes from the vast chemical space. Hatakeyama-Sato et al. [142] constructed the largest lithium-conducting solid polymer electrolyte database so far, and used the GNN and transfer learning for conductivity prediction. This study found that the superionic

conductor composed of charge transfer complexes of aromatic polymers enables a room temperature ionic conductivity of about  $1 \text{ mS cm}^{-1}$ . These materials have much higher glass transition temperatures than PEO, suggesting a novel  $\text{Li}^+$  conduction mechanism in which ion conduction does not depend on the movement of polymer chains. This discovery provides the possibility to break through the glassy design concept of traditional polymer electrolytes. Moreover, Hatakeyama-Sato et al. [143] constructed a new database containing 240 types of lithium-ion conducting solid polymer electrolytes. They extracted 1825 scalar features for the polymer repeat units using the Mordred descriptor. The study employed a gradient boosting model to predict ionic conductivity. The model achieved a determination coefficient  $R^2$  of 0.81 when predicting the test set data. The analysis results showed that the electronegativity and polarity of the monomer units contributed more than 30% to the conductivity, acting as the dominant parameters determining the performance. Inspired by the ML analysis results, the researchers synthesized and characterized new polyglycerol ether derivatives. These functionalized polyethers exhibited superior ionic conductivity at room temperature compared to traditional frameworks, and the experimental results were consistent with the model predictions. Additionally, they used a de novo design algorithm to generate and screen over 15,000 candidate single-ion conductive polymers. Through Bayesian optimization, approximately 200 potential compounds with predicted conductivity reaching  $10^{-4} \text{ S cm}^{-1}$  were finally selected. This study demonstrated that, with a small sample size, choosing the appropriate ML model and descriptors can effectively accelerate the discovery of new materials. Li et al. [144] proposed a workflow integrated DFT calculation and GCNN for screening optimal ionic liquid electrolytes. They collected 2220 ionic liquids as the original dataset, and generated 60 molecule descriptors and 14 electronic structures. Based on unsupervised learning, regression, and classification, they identified 49 valuable candidates and prepared high-performance ionic liquid electrolytes with high ionic conductivity and wide ESW.

Tailoring the key index and developing a robust polymer representation method are necessary. The DN also plays a significant role in regulating  $\text{Li}^+$  transfer in SPE. Gao et al. [140] established a workflow for the prediction of the DN of a polymer. They found that the ion-molecule complex model provides additional correlation analysis between DN and its intrinsic connection with ion-molecule coordination ability. Specifically, HOMO and the binding energy of ion-molecule exhibit a linear relationship with DN value. Guided by this, a high DN zwitterionic polymer electrolyte was designed to promote the uniform ion distribution and  $\text{Li}^+$  transport (Figure 10C). Han et al. [145] proposed a hierarchical polymer graph (HPG) representation method. This method describes the polymer structure at multiple scales, including chemical features at the monomer level and structural features at the chain level. They combined HPG with the graph attention network (GAT) to establish the HPG-GAT deep learning model. This model improves the accuracy and generalization of predicting ionic conductivity. It can handle various structures such as homopolymers, block copolymers, alternating copolymers, and branched polymers. Compared to the traditional monomer molecular graph, HPG-GAT achieved smaller errors and higher determination coefficients on the test set.



**FIGURE 10** | ML for SPE development. (A) Architecture of coarse-grained molecular dynamics–Bayesian optimization (CGMD-BO) framework [138]. Reproduced with permission: Copyright 2020, American Chemical Society [138]. (B) Two-dimensional projection of the space of predicted polymers [139]. Reproduced under the terms of the CC-BY license (<http://creativecommons.org/licenses/by/4.0/>). Copyright 2023, The Authors, published by American Chemical Society [139]. (C) Pearson correlation analysis, heat map, and Interactive mantel tests analysis of molecular features [140]. Reproduced with permission: Copyright 2024, John Wiley-VCH [140]. (D) Workflow of automated-SPE development [141]. Reproduced with permission: Copyright 2025, Elsevier [141].

Quasi-solid polymer electrolyte is an important route that enables high ionic conductivity and interphase compatibility. Zhu et al. [146] proposed an ML-assisted gel electrolyte development method for Zn metal batteries. They combined NN and a training scheme, iteratively inputting modified training data to achieve high predictive accuracy of impedance and corrosion rate. Guided by the model results, a highly cross-linked polyacrylamide (PAM) polymer network was designed. By precisely controlling the swelling time of this polymer in the  $\text{Zn}(\text{ClO}_4)_2$  electrolyte, a “polymer encapsulating water” structure was fabricated. This achieved high electrochemical performance and inhibited the corrosion activity of water on Zn metal and the hydrogen evolution reaction. To improve the electrochemical performance of a solvent-free quasi-solid polymer electrolyte for a lithium metal battery, Lou et al. [147] utilized HTVS and PCA analysis to filter Lithium stearate from 58,680 Li-containing materials. They verified that it possesses a high  $\text{Li}^+$  transfer number and conductivity.

The establishment of the automated laboratory platform can efficiently accelerate the discovery of ML-driven polymer materials. Yang et al. [148] developed a closed-loop pipeline for automatically discovering the formulation of solid polymer electrolytes. They combined a message-passing neural network named ChemArr with Bayesian optimization. This model utilized existing literature

data to accelerate the search process, covering over 1200 possible formulations. The research focused on the combination of poly( $\epsilon$ -caprolactone) (PCL) matrix with 18 different lithium salts. Experimental sample preparation and characterization were conducted through an improved automatic high-throughput platform. In just five optimization batches, researchers discovered a new LiFSI-PCL formulation. This electrolyte achieved an ionic conductivity of  $1.1 \times 10^{-5} \text{ S cm}^{-1}$  at  $30^\circ\text{C}$ , comparable to the performance of standard PEO electrolytes in the field. Moreover, this study confirmed that ionic conduction in the PCL system follows the Meyer–Neldel rule, indicating that ion migration depends on the segmental motion of the polymer chains. Yang et al. [141] developed an automated platform integrated with ML for polymer electrolytes development. Researchers can acquire over 60 SPE samples per hour. All-automated preparation, characterization, and electrochemical measurement could be conducted relying on this workflow. And the feedback of the data to the ML model creates the closed-loop workflow for SPE discovery of sodium battery (Figure 10D).

### 5.3 | ML for Inorganic Solid-State Electrolytes

ISEs are considered the key materials to achieve high performance and high safety all-solid-state batteries due to their

high ionic conductivity, excellent thermal and electrochemical stability, and non-flammability [149]. However, the material selection and optimization of ISE face challenges of broad chemical space, complex multiphysical field coupling, and great difficulty in experimental characterization [150, 151]. ML is able to learn patterns from complex data, build predictive models, and accelerate the material design process to effectively address these challenges in ISE.

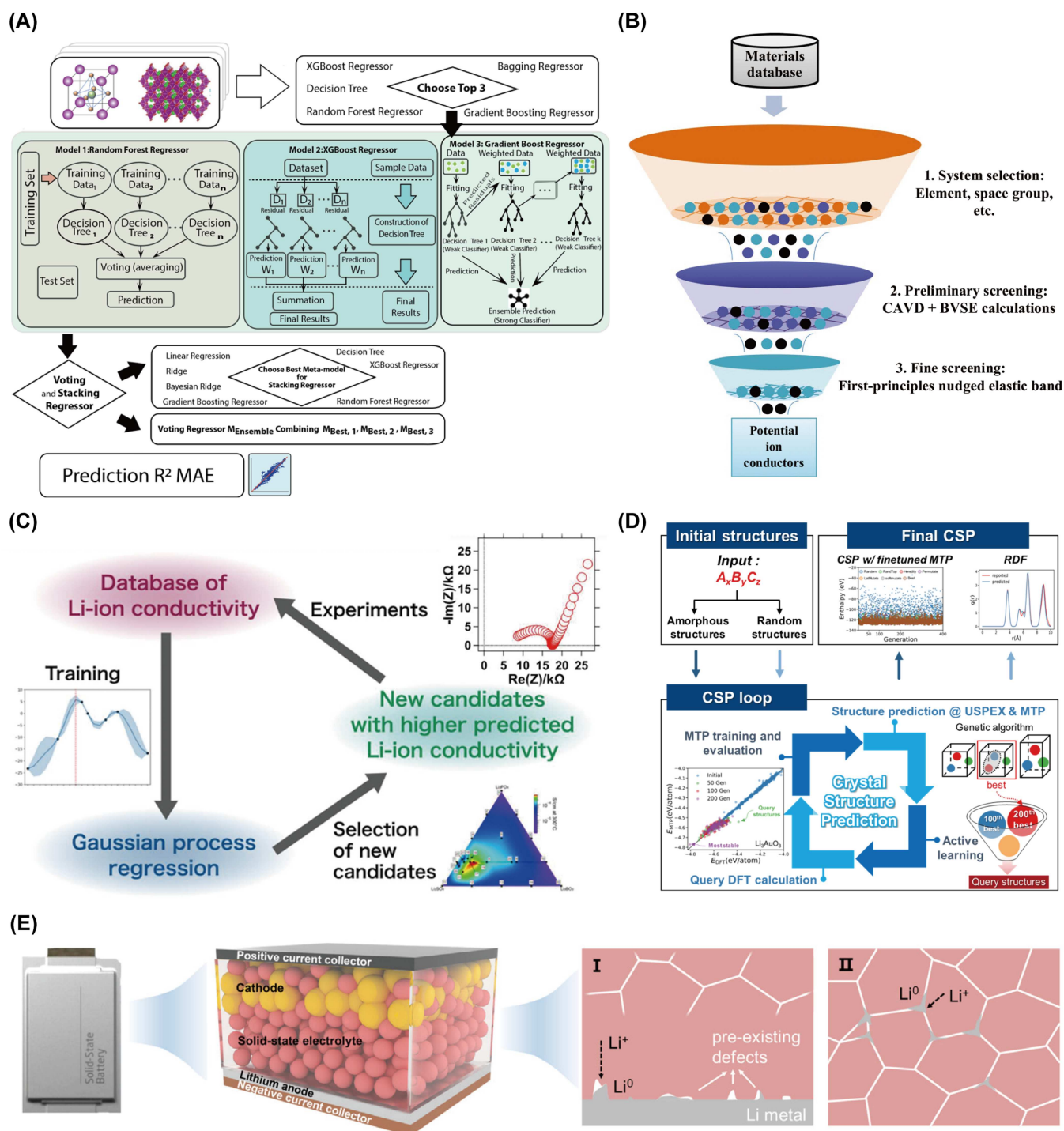
Ionic conductivity is one of the core performance indicators of ISEs. ML provides an effective way for rapid evaluation and optimization of this key performance. In general, by integrating experimental measurements with first-principles calculation data, the chemical composition, crystal structure parameters, thermodynamic stability, ion migration activation energy, and corresponding ionic conductivity of ISEs are systematically collected and converted into high-dimensional descriptors that can be identified by the model [152, 153]. On this basis, many ML algorithms were used to establish the relationship between material descriptors and ionic conductivity. The trained model can be further applied to large-scale material databases such as Materials Project to achieve high-throughput screening and composition optimization, thus significantly accelerating the discovery of new solid electrolytes. Chen et al. [154] used an ML-assisted hierarchical screening strategy to explore the valuable ISE candidates from a dataset containing 20,717 Li-related materials. Firstly, they collected 468 samples from reported literature and trained a ML model for Li-ion conductivity prediction. Combined thermodynamic stability, band gap, and electrochemical stability analysis, three potential candidates were identified from the original dataset. Anass et al. [155] constructed a dataset covering 12,000 DFT calculations of halide ISE for ion conductivity and binding energy prediction. After screening plenty of trivalent and divalent dopants, they confirmed that moderate doping concentration generally yields higher ionic conductivity, while excessive doping tends to deteriorate transport properties. Particularly,  $\text{Ti}^{3+}$  and  $\text{Sc}^{3+}$  trivalent dopants and  $\text{Ra}^{2+}$  and  $\text{Sn}^{2+}$  divalent dopants exhibited the most negative binding energies, which indicates strong stability. In addition, metal-organic frameworks (MOFs) are attracting the attention of battery researchers as they provide 3D porous transportation for  $\text{Li}^+$ . Zhang et al. [156] integrated LLM and ML to discover MOF materials for solid-state electrolyte from 11,393 candidates. LLM was utilized to mine data from reported literature, while ML was used to discover the hidden rules of  $\text{Li}^+$ -MOF interaction. They proposed key design rules about metal centers, framework flexibility, and specific surface area. Guided by this, MOF material NOTT-400 was developed and exhibited high ion conductivity and a wide electrochemical stability window.

Paul et al. [157] collected 3833 data points from reported literature covering 131 materials. They developed the LiCondAI integrated XGBoost, gradient boosting, and RF algorithms to achieve 97% prediction accuracy of ionic conductivity, which greatly improves the screening efficiency of electrolytes for lithium-ion and lithium-metal solid-state batteries (Figure 11A). To address the bottlenecks of low computational efficiency for ion transport properties in solid electrolytes, He et al. [158] proposed a high-throughput screening platform named SPSE. This platform integrates a hierarchical algorithm combining geometric structure analysis (CAVD) and bond-valent site

energy (BVSE) calculations. It enables automated prediction of ion migration pathways and energy barriers within crystal structures, thereby guiding first-principles functional-projected elastic band (FP-NEB) calculations. This work established an open-source platform integrating a materials database, automated computational tools, and ML modules. (Figure 11B). Kenji et al. [159] employed Bayesian optimization to find the optimum composition ratio of the  $\text{Li}_3\text{PO}_3$ - $\text{Li}_3\text{BO}_3$ - $\text{Li}_2\text{SO}_3$  ternary system. They concluded that the ratio at 25:14:61 mol% showed the highest Li-ion conductivity  $4.9 \times 10^{-4} \text{ S cm}^{-1}$ , which is three times higher than the maximum value in the former binary system. Moreover, they used X-ray structure analysis technology and indicated that the formation of a new phase is the key contributor (Figure 11C). Glass-based electrolytes with long-range order and short-range disorder properties act as an important route for ISEs. However, the low ionic conductivity and complex multiple-component space are huge. Hence, establishing a relationship between element candidates and ionic conductivity is meaningful. Different from solely data-driven algorithms, the physics-informed machine learning model (PIML) can integrate knowledge restrictions like physics formulation. Mandal et al. [162] prepared a dataset containing 51 glass compositions of  $\text{Na}_2\text{O}$ - $\text{P}_2\text{O}_5$ - $\text{AlF}_3$ - $\text{SO}_3$  family relying on the melt-quench technique. The Arrhenius equation was added in the PIML and achieved a remarkable 99% fit for ion conductivity, exceeding the conventional ANN model. After that, SHAP analysis found that an increased oxygen element ratio has a positive effect on ionic conductivity. A decreasing sulfur ratio is beneficial for activation energy.

Ab initio molecular dynamics (AIMD) calculation is usually used to investigate the mechanism of ion transport and interfacial reactions at the atomic scale. However, its high computational cost—requiring DFT calculation at each timestep—makes it difficult to cover large systems and long-time scale processes (e.g., reaching nanoseconds or beyond). ML interatomic potentials (MLIPs) or force fields (MLFFs) address this limitation by learning the potential energy surface from DFT data, achieving a significant improvement in computational efficiency while maintaining near-first-principles accuracy. Specifically, an accelerated on-the-fly learning (AOFL) approach was recently demonstrated to track sodium metal-electrolyte reactions, performing 71% faster than AIMD and successfully revealing early-stage SEI formation mechanisms that were previously inaccessible due to the timescale gap [163]. Previous studies have shown that MLIPs can successfully simulate the diffusion behavior of lithium ions in solid electrolytes of  $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$  (LGPS), accurately predict the diffusion rate ( $12 \text{ mS cm}^{-1}$ ) and activation energy (226 meV) at 300 K, and further reveal the regulation of chloride doping on ionic conductivity, which is difficult to be systematically realized under the traditional DFT framework [164, 165]. In addition, the ML is also used to study the diffusion mechanism of fluorine ions in  $\text{BaSnF}$ , which solves the limitation of empirical potentials in accuracy. The temperature-induced phase transition and ionic conductivity evolution in  $\text{Li}_2\text{B}_{12}\text{H}_{12}$  have been successfully simulated [166]. The structural phase transition and fast ion conduction mechanism driven by sodium vacancies in W-doped  $\text{Na}_3\text{SbS}_4$  are revealed [167].

ML also shows attractive ability in material descriptor mining and physical mechanism analysis. By automatically extracting



**FIGURE 11** | ML for ISE development. (A) Architecture of LiCondAI [157]. Reproduced with permission: Copyright 2025, American Chemical Society [157]. (B) Screening process for solid electrolytes based on hierarchical ion-transport algorithm [158]. Reproduced under the terms of the CC-BY license (<http://creativecommons.org/licenses/by/4.0/>). Copyright 2020, The Authors, published by Springer Nature [158]. (C) Workflow of Bayesian optimization for Li-ion conductivity [159]. Reproduced with permission: Copyright 2020, American Chemical Society [159]. (D) Workflow of MTP and active learning [160]. Reproduced with permission: Copyright 2025, American Chemical Society [160]. (E) Schematic of a solid-state lithium metal battery and interface failure mechanisms [161]. Reproduced with permission: Copyright 2025, Elsevier [161].

key structural and chemical features from large-scale data, ML reveals the deep physical and chemical laws behind material properties and provides theoretical guidance for rational material design. The commonly used descriptors include

stoichiometric composition, crystal structure parameters, electronic properties, and defect-related information. With the help of ML methods, the statistical correlation between descriptors and ISEs' performance can be systematically analyzed. And the

relationships between key features and performance indicators can be established through model interpretability analysis tools [168, 169]. For example, in the NASICON-type solid electrolyte system, ML studies have found that sodium stoichiometry is the core factor determining ionic conductivity [170]. At the same time, the neural network model reveals the importance of synthetic parameters and structural factors in regulating conductivity, while the electronegativity of doping elements is relatively weak, highlighting the dominant role of the size and number of dopants. In the cubic Li-argyrodite system, the anion size and its induced structural disorder have been proven to have a significant effect on the activation energy of ion migration. Related studies have proposed a hierarchically encoded crystal structure descriptor (HECS) and achieved high-precision prediction of activation energy through partial least squares analysis [171]. Kim et al. [160] concentrated on sulfide solid-state electrolytes and developed a crystal structure prediction method guided by phase diagrams and ML potential functions. Combining evolutionary algorithms with ML moment tensor potential (MTP) enables the prediction of crystal structures and the relaxation of structures at a lower cost. Subsequently, an active learning strategy was adopted to construct and train the data set. Only a small number of highly uncertain structures were subjected to DFT calculations, and the prediction accuracy of MTP was continuously improved through iterative cycles. They found that edge-sharing topologies significantly reduced the dead volume, providing a larger pathway for  $\text{Li}^+$  diffusion than corner-sharing (Figure 11D). Apart from materials investigation, utilizing ML on the characterization of all-solid-state batteries is important. Li dendrites often nucleate within defects location in the solid-state electrolytes. Wu et al. [161] proposed an efficient ML method for investigating the complicated stress evolution inside the electrolyte (Figure 11E). This method integrates prior knowledge and physics-informed constraints to predict the von Mises stress distribution caused by internal defects. Thanks to the physics-informed constraint model with prior-knowledge incorporated, they achieved facile prediction of stress fields, mechanical degradation, displacement, and failure behavior in the solid-state electrolyte system.

## 6 | Summary and Prospects

Exploring advanced electrolyte materials for next-generation batteries is a key research direction in the energy storage field. As the cornerstone of energy density, rate performance, cycling lifespan, and operating temperature, electrolytes determine ion transfer properties in both the bulk and interphases. Moreover, solid-state electrolytes enable the possibility of all-solid-state batteries with intrinsic safety, which is expected to drive a new revolution in battery science.

AI integrates with almost all research fields, bridging the gap between empirical trials and data-driven rational design. Combining AI with battery research is extremely meaningful, as the development of AI relies on a massive energy supply. Now, ML paradigms have shortened the cycle of materials discovery and reduced associated costs. By knowledge discovery via interpretable models and high-throughput virtual screening via black-box models, ML provides clear guidelines for innovating novel electrolyte materials. While AI technology has

made an impressive impact on electrolyte development, there remain several limitations along with opportunities.

### 6.1 | Data Level: Data Scarcity, and Quality Standardization

AI technology follows the principle of “Garbage In, Garbage Out.” Unlike the massive datasets available in computer vision and natural language processing, the cost of data acquisition for battery testing is high. Furthermore, most electrolyte data reported in the literature are biased, as researchers usually report only optimal battery test results. The lack of large-scale, open-source electrolyte datasets and standardized data reporting systems leads to a scarcity of high-quality data in the electrolyte domain. Therefore, establishing an open-science platform and standardized data reporting protocols is necessary to guarantee data purity, which is essential for the robustness of ML models.

### 6.2 | Encoding Level: Electrolyte Overall Encoding Method

Electrolytes are complex multicomponent systems. Liquid electrolytes contain solvents, co-solvents, salts, and additives, while polymer electrolytes consist of polymer skeletons, salts, and plasticizers. Despite the emergence of several encoding strategies for liquid electrolytes as mentioned above, these methods remain in their infancy. Most existing approaches often struggle to fully capture the nonlinear synergetic effects and intermolecular interactions within such complex multicomponent systems. Therefore, further efforts to develop more holistic and sophisticated representation frameworks are imperative to enable efficient electrolyte-level high-throughput screening.

### 6.3 | Algorithm Level: Knowledge-Data Dual-Driven Model Establishment

Pure data-driven ML models lack physical and chemical constraints, which may lead to unphysical predictions and poor interpretability. For electrolytes, complex molecular interactions and structure–property relationships increase the difficulty of model development. Moreover, the vast background knowledge required across electrochemistry, materials science, physical chemistry, and characterization science poses a challenge for algorithm developers. Therefore, integrating physics-informed ML shows great promise but relies on the collaboration of interdisciplinary teams.

### 6.4 | Experiment Level: Robot-Driven Closed-Loop Laboratory

To generate large-scale, high-quality data through high-throughput experiments, robot-guided automated experimental platforms are necessary. By coupling ML with experimental robots, AI plays the role of both the brain and hands of researchers, achieving the closed-loop autonomous discovery of novel electrolyte materials. In the future, implementing active

learning loops, where models selectively choose the most valuable experimental samples, will exponentially accelerate electrolyte discovery and reduce costs.

Building an intelligent, clean, and green future is a meaningful goal. AI and energy technologies have emerged as critical areas of focus. Among these, ML-driven electrolyte research will significantly accelerate breakthroughs in energy and materials chemistry. Advances in battery technology can provide sustainable energy to support AI development. These two fields synergistically evolve, mutually empowering one another.

## Acknowledgments

The authors acknowledge the support from the Scientific Research Innovation Capability Support Project for Young Faculty (ZYGXQNJSKYCXNLZCXM-M4), National Outstanding Youth Science Fund (52222314), CNPC Innovation Fund (2021DQ02-1001), Dalian Outstanding Youth Science and Technology Talent Project (2023RJ006), and Dalian Science and Technology Innovation Project (2022JJ12GX022).

## Conflicts of Interest

The authors declare no conflicts of interest.

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